

Western Downs Digital Park

Town Planning Report

August 2026

Prepared for: WDDP Pty Ltd



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The river is the symbol of the Dreaming and the journey of life. The circles and lines represent people meeting and connections across time and space. When we are working in different places, we can still be connected and work towards the same goal.

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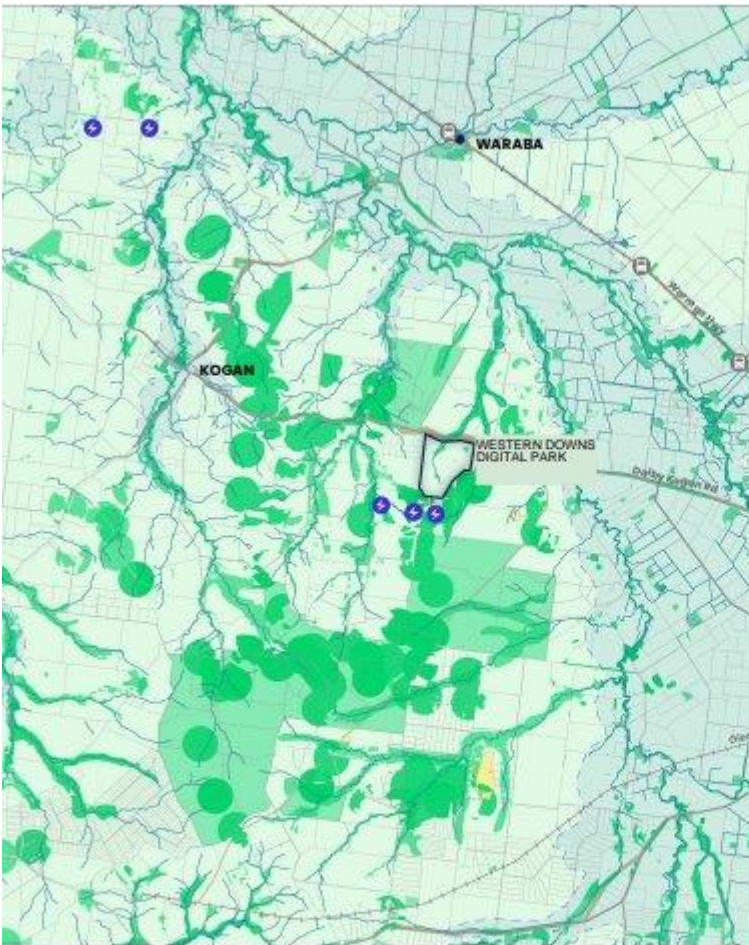


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Executive Summary

The Western Downs Digital Park proposes the establishment of a nationally significant Research and Technology Industry campus at 1933 Dalby–Kogan Road, Dalby, within the Western Downs Region. At full build-out, the development will provide significant digital computing capacity, supported by associated electrical infrastructure, battery-backed power conditioning systems, construction areas, internal access networks and temporary workforce accommodation for up to 1,500 construction workers. The proposal represents a major investment in digital infrastructure and is intended to position Queensland and the Western Downs Region at the forefront of Australia's evolving digital economy.

The site comprises approximately 725.5 hectares of freehold land formally described as Lot 125 on DY316. It is currently partially used for rural purposes and contains the approved Wambo Feedlot. The site is located approximately 37 kilometres north-west of Dalby Town Centre and benefits from direct frontage to Dalby–Kogan Road and Grahams Road. The locality is characterised by a mix of rural, energy and industrial infrastructure uses, including nearby power stations, solar generation, substations, transmission infrastructure and gas infrastructure. These attributes, together with the site's significant land area and separation from urban communities, support its suitability for a large-scale Research and Technology Industry.

The proposed development is subject to Impact Assessment under the *Planning Act 2016* and the *Western Downs Planning Scheme 2017*. Although the land is within the Rural Zone, it is also identified within the Kogan East Special Industry Area, which is intended to accommodate major industrial and special industry activities requiring separation from sensitive land uses. The proposal is therefore supported by strong strategic planning merit, including its proximity to major electricity generation and transmission infrastructure, its ability to accommodate a staged campus within a single large landholding, and its capacity to deliver a regionally significant economic and infrastructure outcome.

The development will comprise four large Research and Technology Industry buildings delivered across four stages, supported by customer-owned switching stations, substations, battery-backed power conditioning systems, access control infrastructure, internal roads, parking, construction laydown areas and a temporary workers' village. Power supply has been planned to leverage the site's proximity to the Braemar generation hub and the broader Queensland transmission network. The project also adopts a water strategy that avoids reliance on town water infrastructure and prioritises air-cooled technology, on-site supply options, rainwater harvesting and treated wastewater reuse to reduce demand on primary water sources.

The planning assessment identifies that the proposal generally responds positively to the strategic direction of the *Western Downs Planning Scheme 2017*. Key planning matters considered in the report include land use compatibility, building height, amenity impacts, strategic location, planning need, community benefit and scenic amenity. While the proposal involves non-rural land uses and exceeds the prescribed building height to a limited extent, the report concludes that these matters are appropriately justified having regard to the site's Special Industry Area designation, substantial setbacks, limited visibility, separation from sensitive receivers and the functional requirements of the development.

A comprehensive suite of specialist reports has been prepared to support the application. These assessments address traffic, transport and access, stormwater management, fire safety, ecology, *Environment Protection and Biodiversity Conservation Act 1999* matters, air quality, noise, external lighting and waste management. The reports generally conclude that the development can be appropriately accommodated on the site, subject to detailed design, recommended mitigation measures and ongoing consultation with relevant authorities. Key measures include upgrades to Dalby–Kogan Road and Grahams Road access arrangements, an integrated stormwater management system, fire safety and containment measures for battery and fuel infrastructure, ecological avoidance and mitigation, lighting controls, waste servicing within the site and further design refinement for air quality and acoustic performance.

The proposal will deliver substantial economic and community benefits. The report identifies approximately 1,500 direct on-site construction jobs during the four-year construction program, approximately 1,400 ongoing direct on-site jobs once the development reaches full operational capacity, and an estimated \$760

million in annual Gross Value Added to the Queensland economy during ongoing operations. The project will also support regional supply chains, attract complementary investment, create opportunities for local businesses and suppliers, broaden employment pathways and strengthen the long-term resilience and diversification of the Western Downs economy.

The proposed engagement strategy seeks to provide transparent, and accessible information to stakeholders and the community throughout the planning process. Engagement will include direct correspondence with adjacent landowners and sensitive receivers, project communications, targeted briefings, stakeholder meetings, and elected representative engagement.

Overall, the report concludes that the Western Downs Digital Park represents a strategically significant development that is appropriately located within an established energy and industrial precinct and can be accommodated without giving rise to unacceptable environmental, infrastructure or amenity impacts. The proposal demonstrates strong strategic and planning merit, responds to a clear need for nationally significant digital infrastructure, and will deliver substantial economic and community benefits to both the Western Downs Region and Queensland. On this basis, the development is considered suitable for approval, subject to appropriate conditions and the continued refinement of technical matters through detailed design and subsequent approval processes.

1 The Proposal

1 The Proposal

1.1 Proposed Development

The proposed development, known as Western Downs Digital Park, comprises a large data centre supported by associated and ancillary infrastructure and land uses. At completion, the data centre will be the largest of its kind in Australia. The development will position Queensland, and the Western Downs region at the forefront of the evolving digital economy.

An outline of the Master Plan for the site is provided in **Figure 1** below.

Figure 1 – Proposed view of the development looking north



Source: HDR

1.2 Description of the proposed land uses

The proposed development consists of the following elements:

- **Four large data centre buildings**, constructed over four development stages (Stages 1A, 1B, 2A and 2B), arranged centrally within the site.
- A **secure operational campus** incorporating controlled access points, boom gates, guard houses, perimeter fencing surrounding the operational areas.
- Components of **associated and ancillary infrastructure** to support an uninterrupted power supply to the proposed development, including:
 - Electrical infrastructure, including substations, switching yards and transmission lines to provide power to the proposed development;
 - Supported by an onsite Battery-Backed Power Conditioning System to strengthen the power grid;
- A **workers accommodation** area which is proposed to provide accommodation for up to 1,500 construction workers during the construction of all stages of the proposed development. In addition to the residential accommodation, areas providing indoor and outdoor recreation and amenities will be provided given the scale of the workforce. It is noted that whilst the location and layout of the

accommodation have been shown on the drawings for the proposed development, there is anticipated to be some changes as a result of future design development;

- A proposed **construction zone**, incorporating a concrete batching plant to be utilised throughout the construction phases of the development. It is anticipated that the footprint of the construction zone will evolve throughout the staged construction delivery.

Figure 2 – Proposed overall site plan



Source: HDR

Vehicular access and servicing of the site is provided from **Dalby-Kogan Road** and **Grahams Road** and includes the following elements:

- boom gates and intercoms;
- guard houses;
- vehicle rejection areas;
- internal circulation roads;
- separate entry and exit arrangements; and
- extensive worker and operational parking.

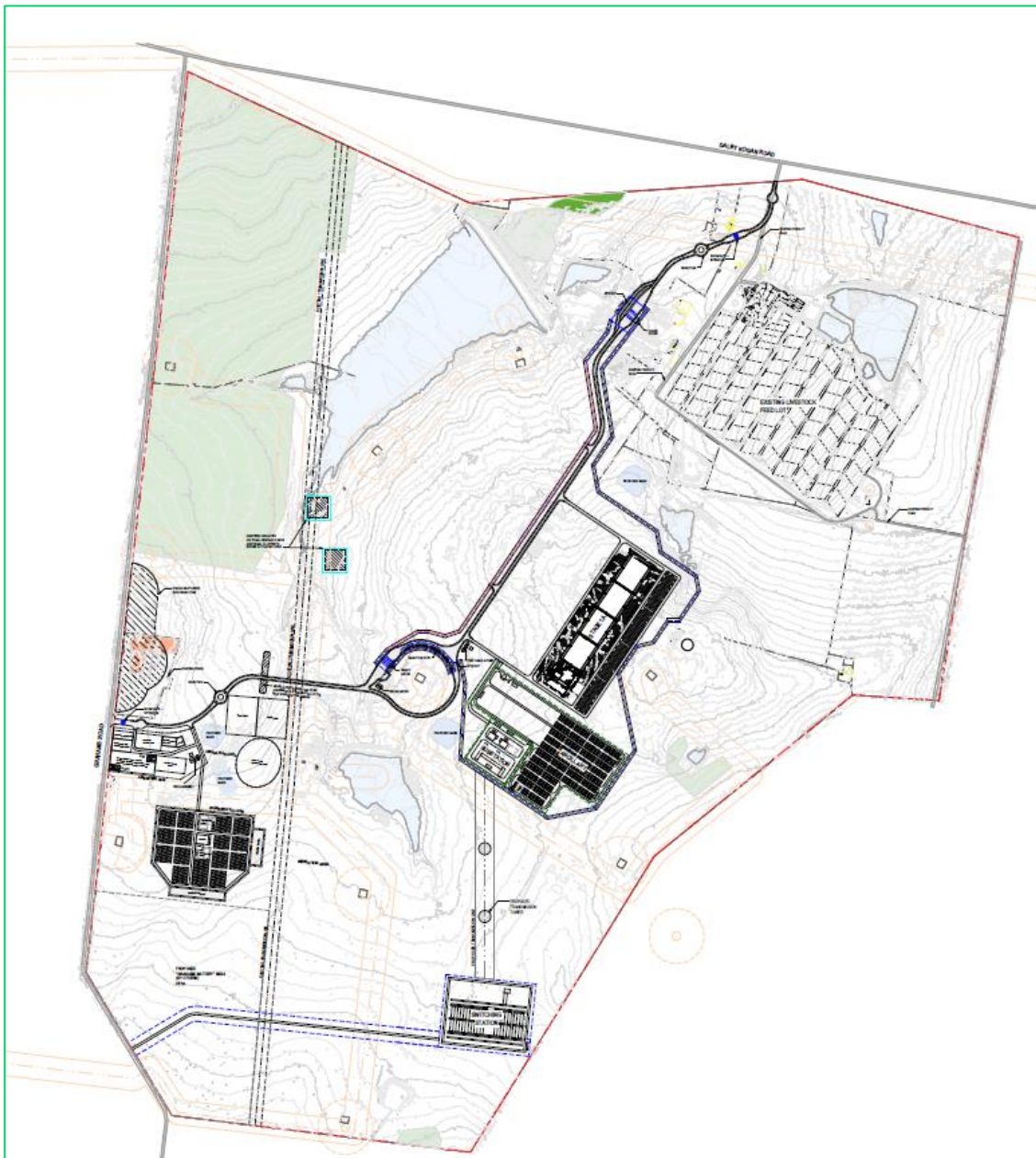
1.2.1 Staging of the development

The proposed development is intended to be delivered across 4 stages as illustrated in the Architectural Drawings included within **Appendix D**. As description of the staged delivery of the development is illustrated below, along with a description of the aspects of each stage.

Stage 1A:

- The Stage 1A data hall building, comprising 36,000sqm of data centre floor space
- Part of the ancillary battery units adjacent to the data hall required to support stage 1A
- Site access from both Grahams Road and Dalby Kogan Road
- The workers village and associated facilities
- The substation and part of the switching station
- Construction lay down areas, including a concrete batching plant which will remain operational for the duration of construction of all stages.

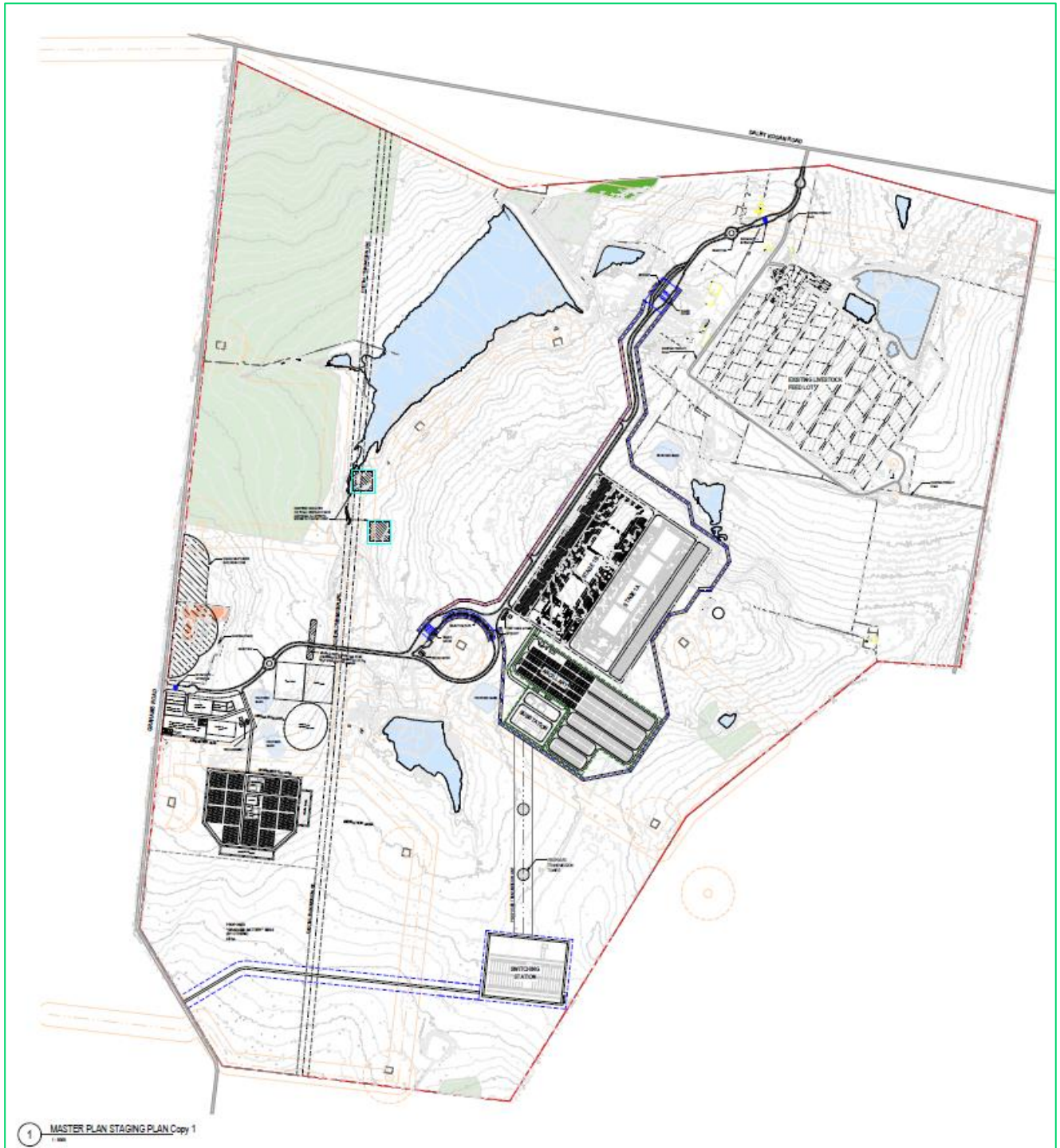
Figure 3 – Stage 1A plan



Stage 1B:

- The Stage 1B data hall building, comprising 36,000sqm of data centre floor space
- Part of the ancillary battery units adjacent to the data hall required to support stage 1B
- The balance of the switching station to deliverer Stages 1A and 1B.

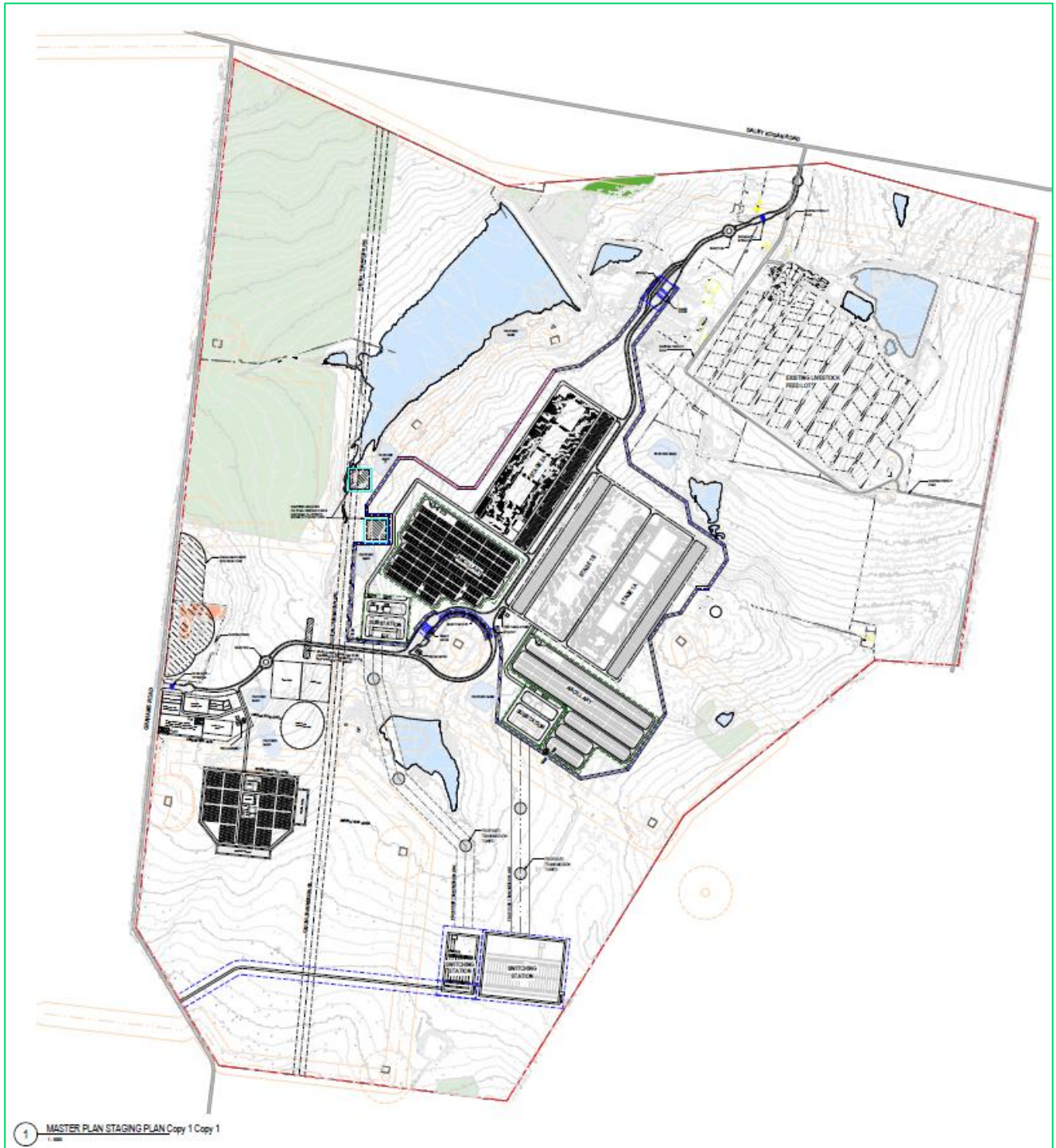
Figure 4 - Stage 1B plan



Stage 2A:

- The Stage 2A data hall building, comprising 36,000sqm of data centre floor space
- Part of the ancillary battery units adjacent to the data hall required to support stage 2A
- The substation and part of the switching station for stage 2A.

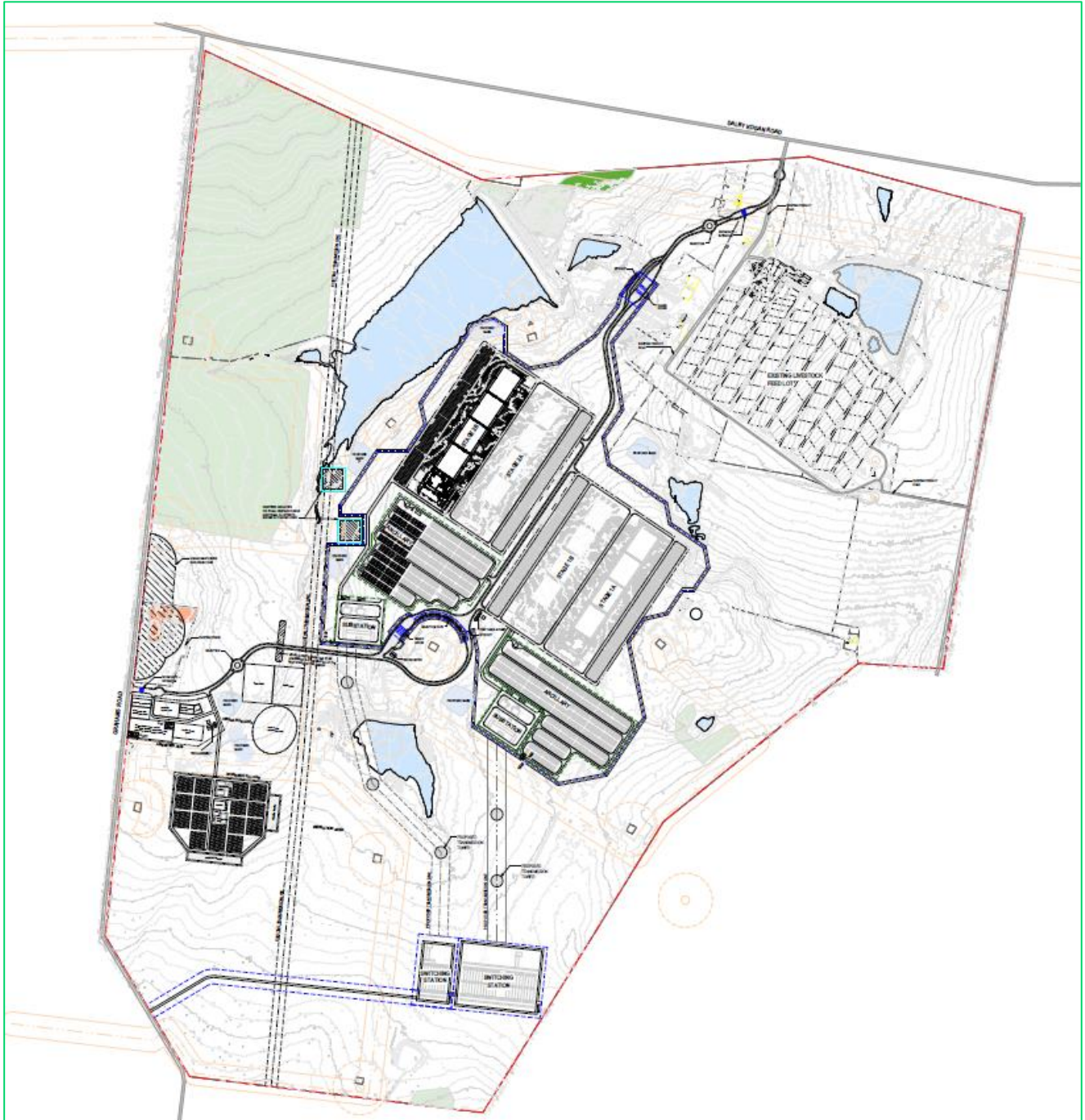
Figure 5 - Stage 2A plan



Stage 2B:

- The Stage 2B data hall building, comprising 36,000sqm of data centre floor space
- Part of the ancillary battery units adjacent to the data hall required to support stage 2B
- The balance of the switching station to deliverer Stages 2A and 2B.

Figure 6 - Stage 2B (ultimate campus) plan



1.3 Operational aspects of the development

The project will be delivered and managed by a consortium of partners with significant experience in developing, operating and managing large-scale, critical infrastructure and digital infrastructure. Central to the project has been consideration of the short- and long-term power and water usage associated with the development. The proposed approach to these key matters is outlined below.

1.3.1 Water usage

The project will not be in competition with the community for water sourcing given the project will utilise primarily air-cooled technology. This eliminates the need for water-based evaporative cooling, significantly reducing water consumption compared to traditional data centres.

There is no reticulated (town) water network in the vicinity of the site, the nearest reticulated water infrastructure being approximately 15km away. The development will therefore be serviced by an independent on-site water supply. This is a well-established servicing arrangement for projects of this nature, and a number of practical and proven supply options are available to meet the development's water requirements for both the construction and operational phases.

Water demand arises from three sources:

- Domestic potable demand (drinking, showering, hand basins and kitchen facilities);
- Domestic non-potable demand (toilet and urinal flushing, washdown, general cleaning and landscape irrigation); and
- Mechanical demand associated with the one-time commissioning (flushing and fill) of the cooling systems.

Demand is highest during construction, with the workers village accommodating a construction workforce of approximately 1,500 persons, and reduces substantially to a modest permanent demand once construction is complete and the village is removed

The development shall be serviced through a combination of the following supply options, either individually or in combination, providing viable pathways to meet the required demand:

- Water delivery – potable water delivered to site by road tanker and held in on-site storage tanks. As the water arrives treated to drinking-water standard, no on-site treatment is required. This provides a readily available supply that can be deployed from the commencement of works and is well suited to the operational demand and to the interim supply during early construction.
- Onsite and offsite recycle water sourcing.
- Surface water from an existing on-site dam, subject to entitlement, reliability and approval.
- Rainwater harvesting and treated wastewater reuse – incorporated as supplementary non-potable sources to reduce demand on the primary supply and improve overall water efficiency.

These options demonstrate that the development can be reliably serviced with potable and non-potable water. The final supply strategy will be confirmed during detailed design, including the mix of primary and supplementary sources, treatment where required, storage and any associated water entitlements, all of which are achievable through established processes.

On-site storage will be provided as an operational buffer against supply interruption. Where water is delivered by cartage, storage is provided as a multi-day buffer at each building, the workers' village and the construction amenities. Where water is drawn from bores or an equivalent continuous source, central treated-water storage provides the main buffer, with a break tank and booster set at each building.

1.3.2 Power Supply

The Western Downs region is a major contributor to the generation of energy in Queensland and a surplus producer of energy. The region hosts major electricity generation assets which has driven the selection of this site as the preferred location for the project. In addition, the Western Downs has become one of Australia's largest renewable energy hubs, with many solar, wind and battery projects exporting power to the broader grid.

The location of the project, proximate to Braemar generation hub and substation provide strong linkage to broader Queensland transmission system. Data centres are increasingly becoming part of the energy landscape in Australia and play a role in converting electricity into digital services, jobs and economic

activity. High quality data centre projects, that leverage technologies and development, can positively impact a region like the Western Downs by allowing it to capture more economic value from its status as an energy capital, rather than exporting most of its energy to other regions.

The location for this project was selected very thoughtfully and strategically to ensure minimal impact on communities and maximum synergy with energy availability, as well as policy and regulatory requirements. The site is directly adjacent to three gas power plants and a solar farm.

The proposal for power sourcing is phased, to ensure upfront access to power that is complementary to the availability via the gas power stations and generators. As part of the project, the construction of a battery storage system, co-located with the Data Centre, will be provided to minimise impact on the network. Notably, the batteries are an ancillary component of the development and are proposed only for the operation of the development and will not have any further role beyond the development.

The campus will be supplied at two voltages i.e., 275kV and 330kV, through two customer-owned switching stations. The first is the 275kV switching station, supplied from two feeders from a power station, two from the Braemar 274kV substation (Powerlink), and five from the Darling Downs Power Station. The second station will operate at 330kV, supplied from the Braemar 330kV substation (Powerlink) via two feeders.

The campus will be delivered in four stages, each establishing a Level 2 substation and the respective data building it supplies, with a stage total facility demand of approximately 540MVA.

- Stage 1A establishes the 275kV switching station with its initial bays, supplied from the power station and Darling Downs Power Station, together with the 275/33kV substation and data building 1A.
- Stage 1B completes the remaining bays of the 275kV switching station and connects them to the Braemar 275kV substation and establishes data building 1B.
- Stage 2 establishes the 330kV switching station, connected to the Braemar 330kV substation and serving the Stage 2 building loads together with the 330/33kV substation and data buildings 2A & 2B.

Each stage is served by a dedicated Level 2 substation stepping the incoming transmission voltage down to 33kV. Each substation comprises three 540MVA transformers. A battery storage system is integrated in series between each 33kV incomer bus and the data building load through a dual power conversion arrangement. The series battery system is multi-purpose: it supports energy arbitrage with the network and allows the campus to present a stable, near-constant load to the Powerlink network.

2

Site and Local Context

2 Site and local context

2.1 Site Location and Tenure

The site is located at 1933 Dalby–Kogan Road, Dalby, and is formally described as Lot 125 on DY316. The site comprises a single freehold parcel of approximately 725.5 hectares (ha) with frontages to Dalby–Kogan Road along the northern boundary and Grahams Road along the eastern boundary. The site is located approximately 37km north-west of Dalby Town Centre.

The site is held under private tenure and is currently uses for rural purposes comprising a cattle feedlot. The site contains an existing approval for a feedlot (known as 'Wambo Feedlot') (Council Ref.: DEV/2004/087). The approved feedlot has a capacity of 24,000 head of cattle. The feedlot and associated ancillary infrastructure are primarily located within the north-eastern portion of the site and include administrative offices, rural worker accommodation and effluent treatment infrastructure.

Figure 7 – Site Aerial



Source: Nearmap, 2025

Existing irrigation areas associated with water reuse and waste treatment are centrally located on the site. The land is predominantly cleared of native vegetation and contains various existing gas well installations distributed across the site. Existing site conditions are depicted in **Figure 8**.

Figure 8 – Site Context



Picture 1 – Western view to the North of the Site

Source: Urbis, 2026



Picture 2 – Western View to the East of the Site

Source: Urbis, 2026



Picture 3 – Southern View to the North of the Site

Source: Urbis, 2026



Picture 4 – Eastern View to the West of the Site

Source: Urbis, 2026

2.1.1 Land Ownership and Encumbrances

Wambo Cattle Company Pty Limited (A.C.N 058 718 326) is the registered owner of the site. A title search confirming land ownership is provided in **Appendix B**. Written consent of the registered landowner to lodge this development application has been provided in **Appendix B**.

The site is subject to two easements, which are summarised below and provided in **Figure 9**.

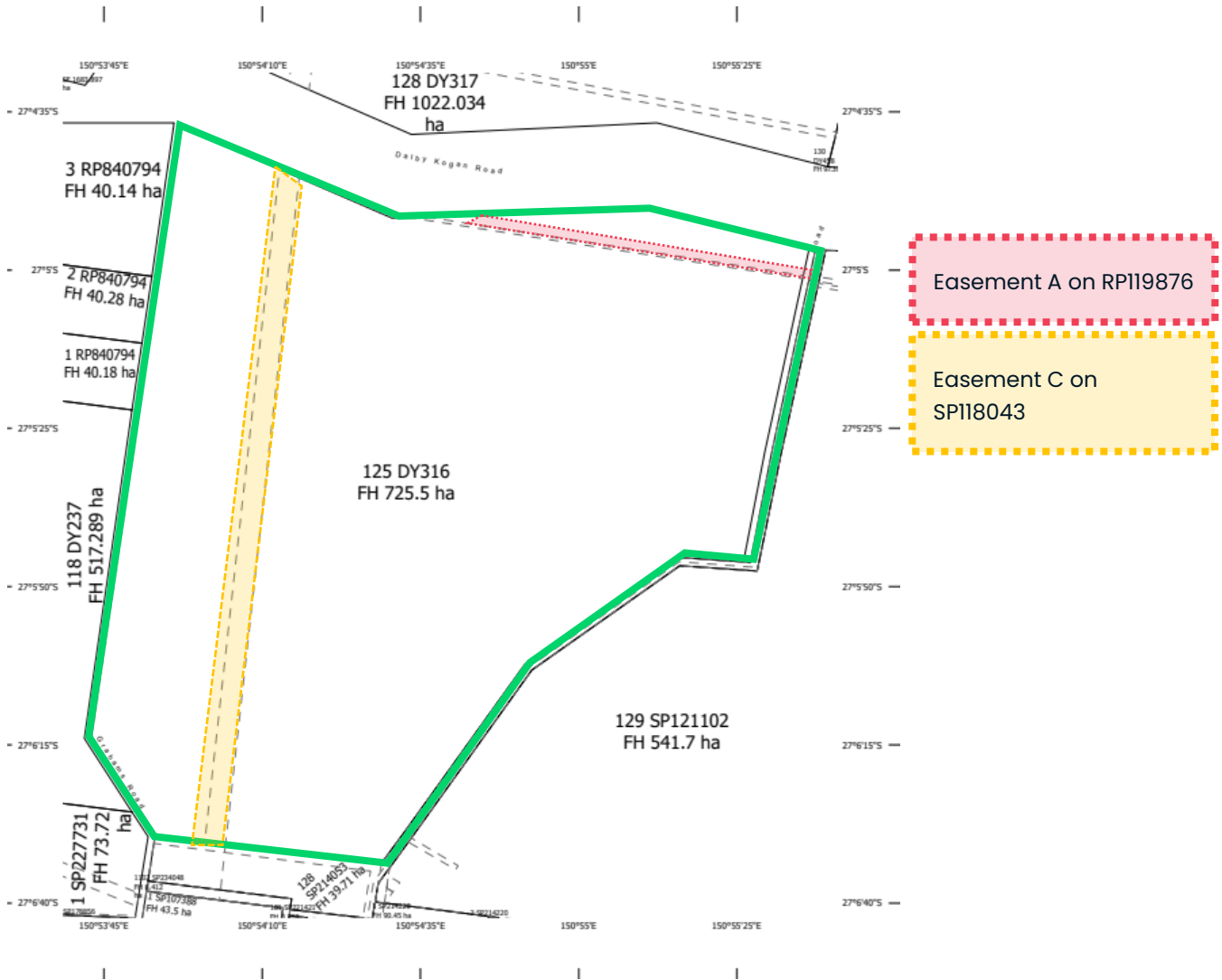
- Easement C on SP118043 – Burdening the site for Electrical Transmission and Electrical Works Purposes;
- Easement A on RP119876 – Burdening the site for Petroleum Pipeline.

Easement and Dealing documents are provided in **Appendix B**.

2.1.2 Cadastral Details

The site comprises a single allotment formally described as Lot 125 on DY316. The site is irregular in shape with a total area of 725.5ha. A cadastral map of the site is provided in **Figure 9**.

Figure 9 – Cadastral Map



Source: Queensland Globe (Marked up by Urbis)

2.2 Surrounding Land Uses and Sensitive Receptors

The surrounding area comprises a mix of rural, industrial and energy infrastructure uses, including battery-backed power conditioning system, power stations and extensive gas well infrastructure. The surrounding locality is characterised by an established rural landholding pattern comprising predominantly smaller rural allotments. The broader locality is predominantly characterised by dryland and irrigated agriculture, grazing and associated rural land uses

Specifically, nearby land uses include the following:

- North** – The site is bounded to the north by Dalby–Kogan Road, a State-controlled road that provides regional connectivity throughout the locality. The site is separated from Dalby–Kogan Road through dense vegetation, providing a visual buffer. Further north are areas associated with Dalby State Forest, together with established gas well infrastructure and associated energy activities – including Dalmore Resources, formally Wilkie Creek mine with operations expected to recommence in 2026. The surrounding land uses reinforce the broader rural and infrastructure-based character of the locality.
- East** – Daandine Power Station is located to the north-east of the site. The facility is owned and operated by APA Group and has an electricity generation capacity of approximately 27.4 Megawatt (MW). The power station is associated with the Daandine and Kogan North Coal Seam Gas Processing

Facilities (CGPFs), which form part of the established energy infrastructure network within the region. Approximately 30.65km east is the Dalby residential area and town centre.

- **South** – The Darling Downs Solar Farm is located to the south-west of the site. The solar farm connects to Powerlink’s Braemar Substation, located south of the subject site. The substation is associated with the Braemar 1 and Braemar 2 Power Stations, which have generation capacities of approximately 504MW and 450MW respectively and are fuelled by natural gas and coal seam gas (CSG). These facilities contribute to the established utility and energy infrastructure character of the broader locality.
- **West** – Land to the west comprises rural landholdings characterised by smaller lot configurations, agricultural uses, and the continuation of the Dalby State Forest. The established rural character are consistent with the proposed development.

Refer to **Figure 10** for further detail regarding the surrounding landholding characteristics.

Figure 10 – Surrounding Context



Picture 5 – View into the site from Dalby-Kogan Road

Source: Google Maps, 2025



Picture 6 – Entry from North to Eastern Informal Access (East)

Source: Google Maps, 2025



Picture 7 – South Entry to Darling Downs Power Station & Solar Farm

Source: Google Maps, 2025



Picture 8 – Aerial from East to adjacent Western Lot

Source: Google Maps, 2025

2.3 Site Access and Transport Connections

Site access is granted through two formal access points, and one unformed access point to the east of the site. Details are as follows:

- **Dalby-Kogan Road** – Dalby-Kogan Road is a state-controlled road located north of the site and is identified as a stock route with a 50m buffer. The site has approximately 2.85km of frontage to the road and provides primary access to the Wambo Feedlot on site.
- **Grahams Road** – Grahams Road is a feeder road located west of the site, with approximately 3.62km of frontage. The road is offset from Dalby-Kogan Road to facilitate safe and efficient site access and provides connectivity to surrounding energy infrastructure, including Braemar Power Stations 1 and 2, and the Darling Downs Power Station and Solar Farm.
- **Eastern Access** – An informal, unnamed access road traverses the north-eastern boundary of the site, with approximately 1.6km of frontage. The road appears to provide access to the existing dwelling on the site and may also facilitate informal access to the Daandine Power Station and associated ancillary infrastructure.

2.4 Site Constraints and Opportunities

Constraints

The proposal has considered a range of environmental, infrastructure and servicing constraints through the detailed design process. While the site is largely cleared, existing wetlands and identified habitat values require careful management to minimise environmental impacts and inform the siting of buildings, infrastructure and access arrangements.

The site is also affected by existing overhead and underground electricity infrastructure associated with the adjoining Powerlink transmission network, together with a telecommunications mast located west of the site along Grahams Road. These constraints have informed the layout and development footprint in **Section 6** and will require ongoing management through detailed design and construction planning.

The site is not currently connected to Council's reticulated water, sewer or electricity infrastructure networks. Accordingly, the proposal incorporates on-site servicing solutions and supporting mitigation measures to address infrastructure requirements. Further detail regarding environmental management, servicing and infrastructure responses is provided in **Section 6**.

Opportunities

The site is a large, consolidated lot with a single-land owner. The site occupies land alongside Dalby-Kogan Road, an arterial road, and is subject to speeds of approximately 100 kilometres per hour. Dalby-Kogan Road provides the single public right of way to the site.

The site benefits from significant mature tree screening along the northern edge of Dalby-Kogan Road, and the eastern Edge of the informal access path to the east of the site, which generally limits views of the site to the north and east. The landholding is significant, allowing for Research and Technology uses to co-locate with residential worker accommodation with an appropriate separation distance. For further details, refer to **Section 2.5** below.

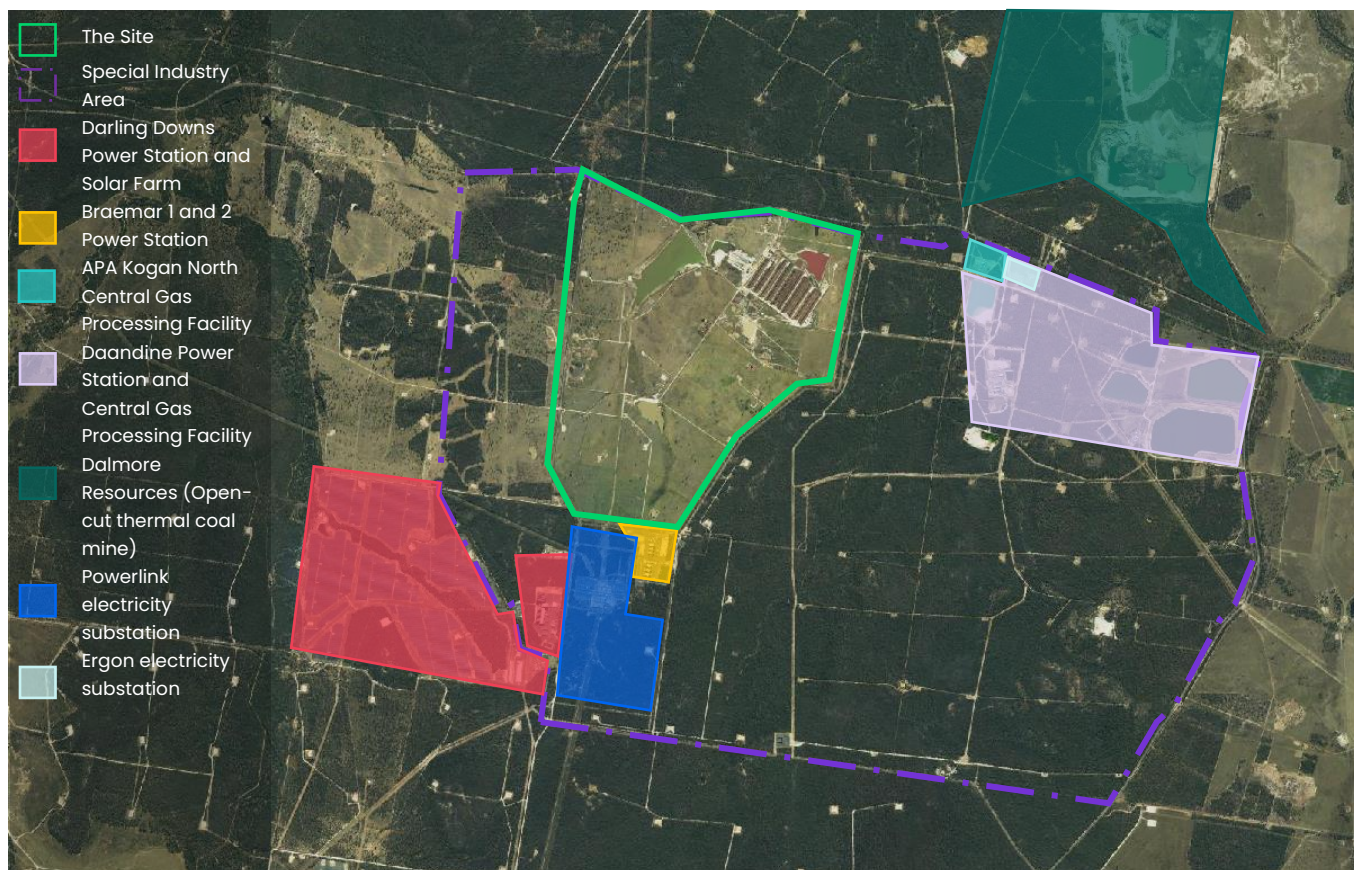
2.5 Special Industry Area and Buffers

The site is identified within the Special Industry Area under Strategic Plan Settlement Pattern Map (SPPM-001) within the Western Downs Regional Council Planning Scheme, supporting the establishment of high impact industry, renewable energy facilities, battery storage facilities and other special industry uses. These areas are intended to accommodate activities requiring separation from sensitive land uses due to the potential for

off-site impacts, including night-time and outdoor operations. Further details to the Strategic Intent of the zoning are provided in **Section 4** of the Town Planning Report.

The surrounding locality contains complementary industrial and energy-related uses, including major electricity infrastructure, the Braemar 1 and 2 Power Stations to the south, and the Darling Downs Power Station and Solar Farm to the south-west of the site. Existing approved uses demonstrate an established development pattern characterised by large-scale industrial activity and appropriate separation from non-industrial uses as seen in **Figure 11**.

Figure 11 – Special Industry Area and Surrounding Uses



Source: Urbis, 2026

2.6 Planning and Approval History

An existing feedlot is approved over the site for the established Wambo Feedlot (Council Ref.: DEV/2004/087). It was approved 9 February 2005 under the *Integrated Planning Act* 1997 and is a Development Permit for a Material Change of Use for an Intensive Animal Feedlot, including an approval for up to a 24,000 head feed lot. The feedlot was granted Environmental Authority (reference: F1-0121 dated 9 July 2013) which authorises the target to conduct ERA 2 – Intensive animal feedlotting (1(c)) – keeping more than 10,000 standard cattle units.

The application currently over the site was approved across three (3) stages as depicted in **Table 1**.

Approved plans are provided in **Figure 12**.

Table 1 – Feedlot Stages

Stage	Development Details
Stage 1	▪ Feedlot contained within one controlled drainage area serviced by a rock weir structure draining to a sedimentation pond with a minimum working capacity of 18.6 Million Litres (ML); ▪ 12 production pens equalling 30,435m²; ▪ 60ML effluent pond.
Stage 2	▪ 26 production pens with a total area of 73,135m²; ▪ Feedlot contained within one controlled drainage area serviced by a rock weir structure draining to a sedimentation pond with a minimum working capacity of 18.6ML; ▪ 60ML effluent pond.
Stage 3	▪ 122 production pens with a total pen area of 365,935m²; ▪ Feedlot contained within two controlled drainage area serviced by a rock weir structure draining to a sedimentation pond with a minimum working capacity of 18.6ML and the other a minimum working capacity of 12.6ML; ▪ Both sedimentation basins serviced by a 99ML effluent pond.

STAGE 2 FEEDLOT LAYOUT

The diagram illustrates a large-scale feedlot facility. It consists of approximately 10 columns of pens. Each pen is numbered sequentially from 1 to over 100. The area is divided by a network of roads and alleys. Key landmarks include:

- Access Roads:** Located along the top and right sides.
- Parking Areas:** Designated spaces for vehicles, primarily along the right side.
- Utilities:** Indicated by lines and symbols throughout the site.
- Orientation:** A north arrow points towards the upper left corner.

Western Downs Digital Park

3

State Planning Matters

3 Statutory Planning Framework

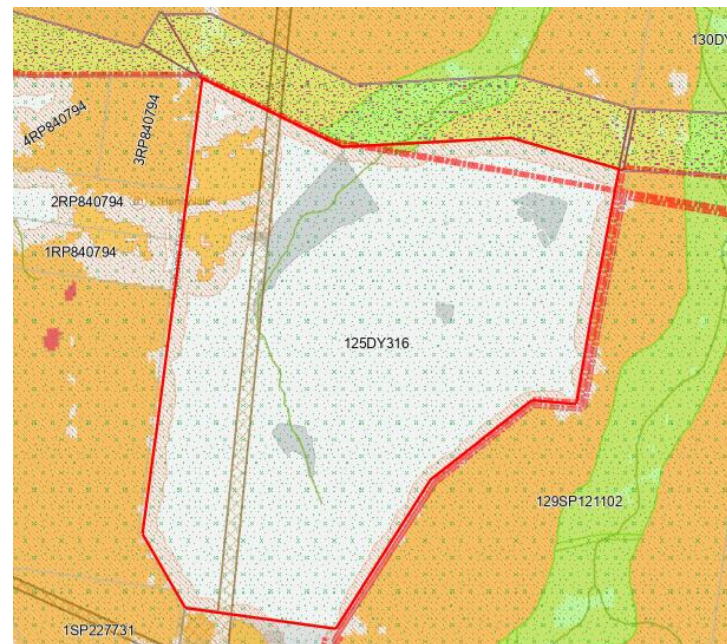
3.1 State Planning Framework Assessment

A summary of compliance with the relevant state planning instruments is outlined in **Table 2** below.

Table 2 – State Planning Framework

Instrument / Assessment Benchmark	Date of Instrument	Assessment
<i>Planning Act 2016</i> (Planning Act)	12 July 2026	The proposed development is subject to the procedures of Impact Assessment and is to be assessed in accordance with Section 45(5) of the Planning Act. Impact assessment must be carried out against the assessment benchmarks in the Planning Scheme and have regard to any matters prescribed by regulation for paragraph 45(5)(b) of the Planning Act.
<i>Planning Regulation 2017</i> (Planning Regulation)	12 July 2026	Schedule 8 of the Planning Regulation identifies the Western Downs Regional Council as the assessment manager for the development application. Further, the development application will require referral in accordance with Schedule 10 of the Planning Regulation.
<i>State Planning Policy</i> (SPP)	3 July 2017	The following state interest matters are relevant to the assessment of this application: ▪ Agriculture: – Stock route network ▪ Biodiversity: – MSES – Regulated vegetation (intersection a watercourse) – MSES – Regulated vegetation (category B) ▪ Emissions and Hazardous Activities: – High pressure gas pipeline ▪ Natural Hazards Risk and Resilience: – Flood hazard area – local government flood mapping area – Bushfire prone area ▪ Energy and Water Supply: – Major electricity infrastructure (Powerlink) ▪ Transport Infrastructure: – State-controlled road The SPP provides supporting mapping to assist in spatially representing policies and requirements contained within the SPP. A comprehensive overlay map of SPP interests is provided in Figure 13 below.

Figure 13 – SPP Mapping Extract



- High pressure gas pipeline
- Major electricity infrastructure (Powerlink)
- MSES – Regulated vegetation (intersecting a watercourse)
- MSES – Regulated vegetation (category B)
- State-controlled road
- Flood hazard area – local government flood mapping area
- Stock route network
- Bushfire prone area
- Very High Potential Bushfire Intensity
- High Potential Bushfire Intensity
- Medium Potential Bushfire Intensity
- Potential Impact Buffer

The applicable Planning Scheme (*Western Downs Planning Scheme 2017*) has been prepared to fully reflect the interests of the SPP. Therefore, the proposed development will not require assessment against the SPP.

<i>Darling Downs Regional Plan 2013</i> (Regional Plan)	15 October 2013	The subject site is located within the Priority Agricultural Area identified under the Regional Plan. The Regional Plan seeks to protect Priority Agricultural Land Uses while recognising opportunities for the co-existence of agriculture, industry and other regionally significant economic activities. The proposed development introduces new land uses that will be assessed through this development application. The proposal has been planned having regard to the strategic intent of the Regional Plan, including the need to balance
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protection of productive agricultural land with opportunities for significant economic investment, employment generation and regional growth. The proposal is considered to respond to the broader economic objectives of the Regional Plan, while recognising the strategic importance of the locality and the existing agricultural context.

It is noted that the Queensland Government is currently undertaking a review of all regional plans across the State to ensure they continue to respond to contemporary legislative, policy and growth priorities. At the time of preparing this report, the *Darling Downs Regional Plan 2013* remains the applicable regional planning instrument.

An extract of the Regional Plan mapping is provided in **Figure 14** below.

Figure 14 - Regional Plan Mapping Extract



State
Development
Assessment
Provisions Version
3.6

(SDAP)

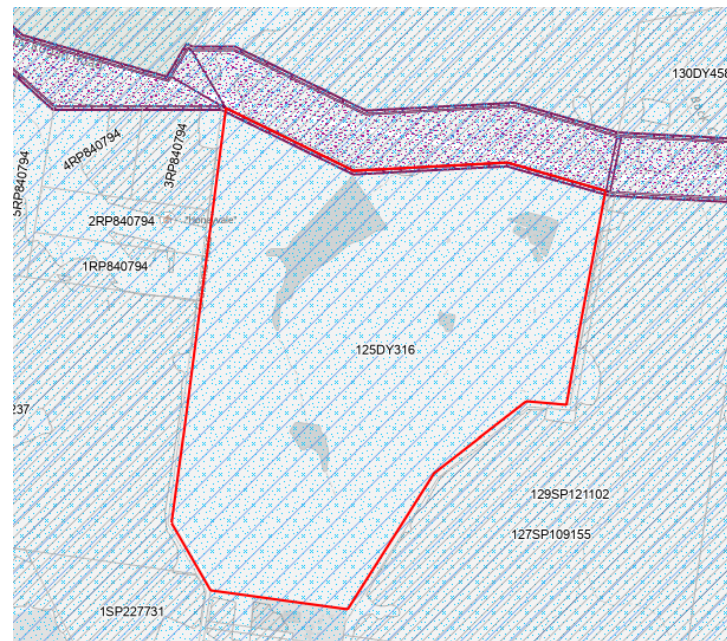
27 April 2026

The mapping indicates the following SDAP matters apply to the site:

- Water Resources:
 - Water Resource Planning Area Boundaries
 - Great Artesian Water Resource Plan Area
- Native Vegetation Clearing:
 - Regulated Vegetation Management Map (Category A And B Extract)
- State Transport Corridor:
 - State-Controlled Road
 - Area Within 25m Of a State-Controlled Road

An extract of the SDAP mapping is shown in **Figure 15**, illustrating the extent of the mapped state interests.

Figure 15 – SDAP Mapping Extract



-  The Site
-  Water resource planning area boundaries
-  Great artesian water resource plan area
-  State-controlled road
-  Area within 25m of a State-controlled road

Refer to **Section 3.2** below regarding the applicable referral and relevant SDAP codes for the proposed development.

3.2 Referral Triggers

The proposal requires referral to the State Assessment and Referral Agency (**SARA**) under Schedule 10 of the Planning Regulation for the matters outlined in **Table 3** below.

Table 3 – Referral Triggers

Planning Regulation Reference	Relevant Assessment Benchmark
Schedule 10, Part 9, Division 4, Subdivision 1, Table 1	Development application for an aspect of development stated in schedule 20 that is assessable development under a local categorising instrument or section 21, if – (a) the development is for a purpose stated in schedule 20, column 1 for the aspect; and (b) 1 the development meets or exceeds the threshold – (i) for development in local government area 1—stated in schedule 20, column 2 for the purpose; or (ii) for development in local government area 2—stated in schedule 20, column 3 for the purpose; and (c) for development in local government area 1—the development is not for an accommodation activity or an office at premises wholly or partly in the excluded area

However, if the development is for a combination of purposes stated in the same item of schedule 20, the threshold is for the combination of purposes and not for each individual purpose.

Schedule 10, Part 9, Division 4, Subdivision 2, Table 4	Development application for a material change of use, other than an excluded material change of use, that is assessable development under a local categorising instrument, if all or part of the premises – (a) are within 25m of a State transport corridor; or (b) are a future State transport corridor; or (c) are – (i) adjacent to a road that intersects with a State-controlled road; and (ii) within 100m of the intersection
Schedule 10, Part 9, Division 2, Table 2	Development application for a material change of use that is assessable development under a local categorising instrument and does not relate to reconfiguring a lot, if– (a) all or part of the premises are within 100m of a substation site; or (b) both of the following apply – (i) all or part of the premises are subject to an easement for the benefit of a distribution entity, or transmission entity, under the Electricity Act; (ii) the easement is for a transmission grid or supply network
Schedule 10, Part 9, Division 3, Table 2	Development application for a material change of use that is assessable development under a local categorising instrument and does not relate to reconfiguring a lot, if– (a) all or part of the premises are subject to an easement for the benefit of the holder of a pipeline licence under the Petroleum and Gas Act; and (b) the easement is for the construction or operation of the pipeline that is the subject of the pipeline licence; and (c) all or part of a structure or work that is the natural and ordinary consequence of the use is, or will be, on the easement

Accordingly, this application will be referred to SARA (DTMR), Powerlink and APT Petroleum Pipelines Pty Ltd (A.C.N. 009 737 393) as concurrence agencies for the relevant referral triggers. As the proposed development required referral to SARA, the SDAP is applicable. An assessment of the SDAP indicates that the development requires assessment against the following benchmarks:

- State Code 1: Development in a State-controlled Road Environment

An assessment against State Code 1 is provided in **Appendix F**.

3.3 Public Notification

In accordance with Section 53 of the Planning Act, the notification stages of the development assessment process apply to an application if either of the following applies –

- (a) Any part of the application requires impact assessment; or
- (b) The application includes a variation request.

As the proposal involves an impact assessment, public notification pursuant to the Planning Act is required in this instance for a minimum of 15 business days.

4 **Local Planning Framework**

4 Local Planning Framework

The *Western Downs Regional Plan 2017 (Version 6)* (the '**Planning Scheme**') is the local planning instrument relevant to the assessment of the proposed development.

The following sections of this report provide an assessment of the proposed development against the relevant assessment benchmarks in the Planning Scheme.

4.1 Strategic Outcomes

As the proposed development is subject to Impact Assessment, the application is required to be assessed against the entire Planning Scheme, including the Strategic Framework contained within Part 3.

The Strategic Framework establishes the policy direction for the Planning Scheme and provides the basis for ensuring appropriate development occurs throughout the planning scheme area over the life of the scheme. The Strategic Framework is structured around five (5) overarching themes that collectively represent the long-term planning vision for the Western Downs Region:

- Liveable Communities and Housing;
- Environment and Heritage;
- Economic Growth;
- Infrastructure; and
- Safety and Resilience to Hazards.

The five themes articulate a series of strategic outcomes, elements, specific outcomes and land use strategies that guide future development across the region. While the Strategic Framework must be considered in its entirety, the proposal is primarily influenced by those themes and elements relating to settlement pattern, industrial development and infrastructure. The remaining themes are addressed, where relevant, through the technical assessments accompanying this application.

4.1.1 Theme 1 – Liveable Communities and Housing

The Liveable Communities and Housing theme establishes the desired settlement pattern for the Western Downs Region and identifies the preferred locations for future urban, industrial and rural development. Of particular relevance, the Strategic Plan Settlement Pattern Map identifies the subject site within the Special Industry Area (Kogan East), recognising the locality as an appropriate location for regionally significant industrial and special industry uses requiring separation from sensitive land uses.

The proposal has been planned having regard to the strategic intent of the Special Industry Area and responds to the Planning Scheme's objective of directing major industrial activities to locations with appropriate separation distances, access to strategic infrastructure and compatibility with surrounding land uses. The proposal also acknowledges the site's rural setting and proximity to existing agricultural activities, recognising the importance of balancing regional economic development with the ongoing operation of established agricultural enterprises.

Having regard to the nature of the proposal, the following elements of Theme 1 are considered to be of particular relevance:

- Element 3.3.2 – Network of Centres; and
- Element 3.3.9 – Industrial Land Use and Development

Table 5 provides an assessment against the relevant land use strategies for Element 3.3.2 – Network of Centres.

Table 4 – Assessment against Specific Outcome of Strategic Framework – Element 3.3.2

Specific Outcome	Assessment of Proposal
SO4 Miles and Wandoan are transport and service hubs for the resource and agricultural sectors. Their accessibility to transport corridors, proximity to emerging resource sector activities and availability of well serviced industrial land is utilised to support the supply chain servicing resource sector operations in the Western Downs and throughout the eastern part of the Surat Basin. Development of an intermodal transport and logistics hub is supported in Miles given its location to transportation routes.	While the subject site is not located within Miles or Wandoan, the proposal supports the broader strategic intent of facilitating regionally significant industrial development within the Western Downs. The site benefits from proximity to the Kogan Industrial Area, existing energy infrastructure, strategic road connections and the wider Surat Basin, supporting continued investment and economic growth within the region.
SO5 Townships provide for limited urban development and serve to meet the daily needs of the surrounding rural district, consistent with the scale and intensity of existing urban activities.	The proposal is located outside the Kogan Township and does not seek to introduce urban development within the township. The scale and nature of the proposed development will not alter the intended role or function of Kogan as a rural township and is therefore consistent with the broader settlement hierarchy established by the Strategic Framework.

Table 6 provides an assessment against the relevant land use strategies for Element 3.3.9 – Industrial Land Use and Development.

Table 5 – Assessment against Specific Outcome of Strategic Framework – Element 3.3.9

Specific Outcome/ Land Use Strategy	Assessment of Proposal
SO1 Western Downs is responsive to the industrial needs of the resource, energy and agricultural sectors where the long-term health of the environment and wellbeing of communities is protected.	The proposal directly responds to this outcome through the establishment of a regionally significant Research and Technology Industry, supported by Workforce Accommodation and High Impact Industry necessary to facilitate the construction and ongoing operation of the development. The proposal represents significant private investment in the Western Downs and has been supported by a comprehensive suite of technical assessments demonstrating that environmental and amenity impacts can be appropriately managed.
SO2 The location of industrial activities avoids or is appropriately separated from sensitive land uses to protect waterway health and the health, wellbeing, amenity and safety of the community	The subject site is strategically located within the Kogan East Special Industry Area and benefits from substantial separation distances to sensitive land uses. The proposal has been designed to respond

from the impacts of air, noise and odour emissions and hazardous materials.

to the site's surrounding context, with specialist technical assessments demonstrating that potential environmental and amenity impacts can be appropriately mitigated.

SO3

Industrial development incorporates best practice environmental performance and design to reduce the potential for noise, particulate and odour emissions, stormwater runoff and wastewater generation.

The proposal incorporates contemporary environmental management measures supported by specialist investigations addressing ecology, stormwater, air quality, noise, waste management and other environmental matters. Collectively, these measures demonstrate that the proposal adopts best practice environmental performance and design outcomes.

SO4

High impact industry and Special Industry uses that require separation distances from sensitive uses and that may have potential to generate off-site impacts are supported in Special Industrial Areas only as identified on Strategic Plan Map 1 – Settlement Pattern.

The subject site is identified within the **Kogan East Special Industry Area** on Strategic Plan Map 1. The proposal therefore directly aligns with the Strategic Framework's intent of accommodating regionally significant industrial development within locations specifically identified for Special Industry uses.

SO5

Renewable energy facilities and battery storage facilities are supported throughout the region where appropriately located in respect to nearby sensitive land uses, where off-site impacts can be appropriately mitigated and the viability of agricultural land is retained.

While the primary use comprises a Research and Technology Industry, the proposal includes ancillary energy infrastructure that capitalises on the site's strategic location adjoining existing electricity generation and transmission infrastructure. The proposal has been planned to integrate with this infrastructure while ensuring potential off-site impacts are appropriately managed.

LUS6

The Special Industrial Areas identified on Strategic Plan Map 1 – Settlement Pattern indicate where future Special Industry development is to be located.

The proposal directly responds to this strategy by locating a Research and Technology Industry within the Kogan East Special Industry Area. The development reinforces the intended strategic function of the precinct and represents the type of major industrial investment envisaged by the Strategic Framework.

LU8

Industrial uses will not be supported where established on Rural land that is considered to be productive agricultural land or of an allotment size that is economically viable for agricultural production. Where development is proposed on such land, overriding community need must be demonstrated together with justification that there is no suitably zoned land available.

The proposal is accompanied by a detailed planning justification demonstrating the strategic need for the development and the suitability of the subject site. Importantly, the proposal responds to the Planning Scheme's identification of the land as a Special Industry Area, while also retaining the existing feedlot operation on a separate allotment, allowing the continued operation of this established agricultural use. The proposal therefore

achieves an appropriate balance between supporting regionally significant economic development and maintaining ongoing agricultural production.

LU9

Where development is not consistent with the purpose and intent of the zone, overriding community need will need to be demonstrated together with valid planning justification as to why the proposed use cannot reasonably be established in a more appropriate zone.

The proposal is supported by detailed planning justification demonstrating significant economic, employment and infrastructure benefits, together with site-specific locational attributes that make the subject land uniquely suited to the proposed development. These include the site's designation within the Kogan East Special Industry Area, direct access to strategic electricity infrastructure, separation from sensitive land uses and ability to accommodate the scale and operational requirements of the proposed Research and Technology Industry. The ancillary Workforce Accommodation is necessary to support the significant construction and operational workforce associated with the development and forms an integral component of the overall proposal.

4.1.2 Theme 3 – Economic Growth

The Economic Growth theme recognises the importance of industrial development in supporting the continued economic growth and diversification of the Western Downs Region. The Strategic Framework encourages industrial investment in locations that are appropriately serviced by infrastructure, protected from incompatible development and capable of supporting the long-term needs of the resource, energy and agricultural sectors.

The proposal aligns with these strategic objectives by facilitating significant private investment in the region, creating substantial employment opportunities during both construction and operation, and supporting the continued diversification of the regional economy. The proposal also responds to the Strategic Framework's objective of accommodating regionally significant industrial development in appropriate locations supported by existing infrastructure and transport networks.

4.1.3 Theme 4 – Infrastructure

The Infrastructure theme promotes the efficient and orderly provision of infrastructure to support growth across the Western Downs Region. The Strategic Framework recognises the importance of coordinating development with major transport, energy, telecommunications and utility infrastructure to ensure infrastructure investment occurs in an efficient and integrated manner.

The proposal has been planned to capitalise on the site's strategic location adjoining existing electricity generation and transmission infrastructure, while also maintaining access to the regional road network and supporting utility services. The proposal therefore responds positively to the Strategic Framework by locating significant industrial development in a manner that optimises existing infrastructure investment and minimises the need for unnecessary expansion of regional infrastructure networks.

Overall, the proposal is considered to respond positively to the strategic direction established by the Western Downs Planning Scheme. It supports the strategic intent of the Kogan East Special Industry Area, contributes to the economic and infrastructure objectives of the region and establishes an appropriate planning framework for assessment against the applicable zone, overlay and development codes addressed throughout the remainder of this report.

4.2 Land Use Definitions

The proposed development involves the following land uses, as defined within the Planning Regulation and illustrated in **Table 7** below.

Table 6 – Land Use Definition

Use	Western Downs Planning Scheme Use Definition
Research and Technology Industry	<i>Use of premises for an innovative or emerging industry that involves designing and researching, assembling, manufacturing, maintaining, storing or testing machinery or equipment.</i>
Workforce Accommodation	<i>Use of premises for:</i> <i>(a) means the use of premises for:</i> <i>(i) accommodation that is provided for persons who perform work as part of:</i> <i>A. a resource extraction project; or</i> <i>B. a project identified in a planning scheme as a major industry or infrastructure project; or</i> <i>C. a rural use; or</i> <i>(ii) recreation and entertainment facilities for persons residing at the premises and their visitors, if the use is ancillary to the use in subparagraphs (i); but</i> <i>(b) does not include rural workers' accommodation.</i>
High-impact Industry	<i>Use of premises for an industrial activity:</i> <i>(a) that is the manufacturing, producing, processing, repairing, altering, recycling, storing, distributing, transferring or treating of products; and</i> <i>(b) that a local planning instrument applying to the premises states is a high impact industry; and</i> <i>(c) that complies with any thresholds for the activity stated in a local planning instrument applying to the premises, including, for example, thresholds relating to the number of products manufactured or the level of emissions produced by the activity.</i> <i>Note: examples of the use include a concrete batching plant and storage of dangerous goods</i>

4.3 Zoning

In accordance with the planning scheme, the site is identified within the Rural zone, as illustrated in **Figure 16** below.

The purpose of the **Rural Zone** is to:

- “(a) provide for rural uses and activities; and*
- (b) provide for other uses and activities that are compatible with:*
 - (i) existing and future rural uses and activities; and*
 - (ii) the character and environmental features of the zone; and*
- (c) maintain the capacity of land for rural uses and activities by protecting and managing significant natural resources and processes.”*

The Rural Zone code is supported by the following overall outcomes:

- “(4) New enterprises are accommodated where they are directly associated with rural production, a natural resource or the natural environment, or require separation from urban uses due to their operational characteristics. Such uses are to maintain the productive capacity of the land, avoid conflicts with surrounding activities, protect landscape and natural resource values, and demonstrate they cannot be more appropriately located within another zone.*
- (6) The environmental, character and landscape values of all rural land are protected from encroachment by incompatible land uses.*
- (8) Special industry uses that require separation distances from sensitive land uses are supported and encouraged to locate in areas identified as Special Industrial Area identified on Strategic Plan Map 1 – Settlement Pattern.*
- (10) Development is connected to available infrastructure networks or is provided with appropriate on-site infrastructure to protect environmental health and human wellbeing.*
- (11) Ecologically significant features, including waterways, wetlands and significant vegetation, are retained and appropriately buffered from the impacts of development.*
- (12) Development is designed to achieve ecological sustainability through energy efficiency, water conservation, water quality management and Crime Prevention Through Environmental Design (CPTED) principles.*
- (14) Development responds appropriately to natural hazards and site constraints and does not increase the exposure of people or property to risk.*
- (15) Where development is not consistent with the purpose and intent of the Rural Zone, overriding community need and planning justification are demonstrated, including why the proposed use cannot reasonably be established within a more appropriate zone.”*

The proposal has been developed having regard to the purpose and intent of the Rural Zone and the strategic planning direction established by the Planning Scheme. While the proposed Research and Technology Industry and associated Workforce Accommodation represent non-rural land uses, the Planning Scheme expressly recognises that other compatible uses may be appropriate within the Rural Zone where they respond to the strategic intent of the locality.

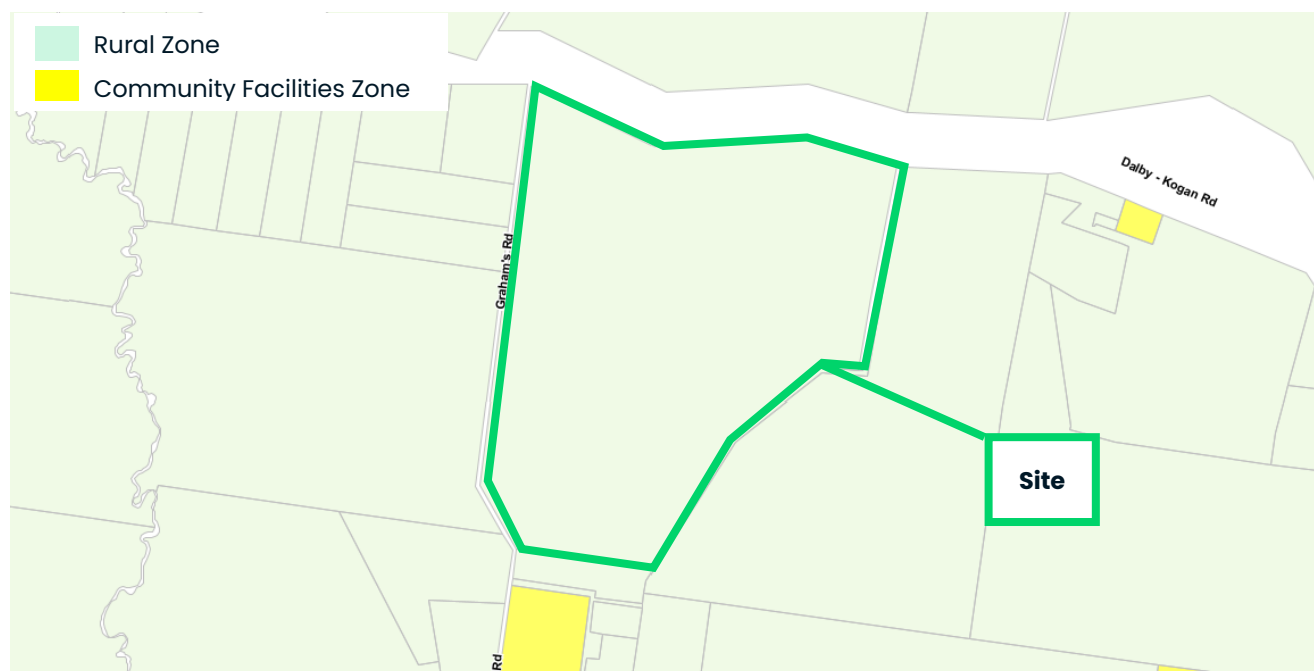
Importantly, the subject site is identified within the Kogan East Special Industry Area, which has been specifically designated to accommodate regionally significant industrial development requiring separation from sensitive land uses. The proposal therefore directly responds to the Planning Scheme's strategic land use framework by locating the proposed development within an area identified for this purpose.

The proposal also benefits from significant separation from sensitive land uses, direct access to major electricity generation and transmission infrastructure, and sufficient land area to accommodate the operational requirements of the development while managing environmental and amenity impacts. Supporting technical assessments demonstrate that the proposal can be appropriately serviced, responds to the site's environmental characteristics and natural hazards, and incorporates suitable mitigation measures where required.

The ancillary Workforce Accommodation has been integrated as a supporting component of the development to accommodate the substantial construction workforce associated with the project. In this regard, it forms an integral part of the overall development rather than a standalone residential use.

Overall, the proposal is considered to achieve the intent of the Rural Zone by delivering a regionally significant development within the Kogan East Special Industry Area, while maintaining compatibility with the surrounding rural environment and demonstrating an appropriate planning outcome having regard to the strategic direction of the Planning Scheme.

Figure 16 – Zoning Map



Source: Western Downs Planning Scheme 2017 (Marked up by Urbis)

4.4 Category of Development and Assessment

4.4.1 Category of Development

There are three categories of development under the Planning Act. These are Accepted Development, Assessable Development and Prohibited Development.

The planning scheme, acting as the Categorising Instrument, determines that the category of development for the proposed development is Assessable Development.

4.4.2 Category of Assessment

The Western Downs Planning Scheme determines the category of assessment for the proposed development through the applicable category of development and assessment tables.

Table 5.5.10 – *Rural Zone* identifies the categories of assessment for Material Change of Use within the Rural Zone. Neither the proposed Research and Technology Industry nor the associated Workforce Accommodation use is identified within the table. Accordingly, both uses default to **Impact Assessment** under the Planning Scheme.

As the application is subject to Impact Assessment, it is required to be assessed against the entire Western Downs Planning Scheme in accordance with the Planning Act. This includes consideration of the Strategic Framework, the Rural Zone Code, applicable overlay codes and all other relevant assessment benchmarks.

The application will also be subject to public notification in accordance with the Development Assessment Rules, enabling public submissions to be made during the statutory notification period.

4.5 Overlays

Parts of the site are identified within the overlays listed in **Table 8** below.

Table 7 – Overlays

Overlays	Sub-category
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Biodiversity Overlay	▪ MSES – Regulated Vegetation
Bushfire Risk Overlay	▪ Medium Potential Bushfire Intensity
Infrastructure Overlay	▪ Pipeline Buffer (Gas Pipeline) ▪ Pipeline Licences (Gas Pipeline) ▪ Transmission Corridor
Stock Routes Overlay	▪ Stock Route (50m Buffer)
Scenic Amenity Overlay	▪ High Landscape Value Areas
Wetlands Overlay	▪ Wetlands (Lacustrine) ▪ Wetlands Trigger Area
Road Hierarchy Overlay	▪ State-controlled road (Dalby-Kogan Road) ▪ Feeder Road (Grahams Road)
Noise Corridor Overlay	▪ Category 0: Noise Level < 58 dB(A) ▪ Category 1: 58 – 62 dB (A) ▪ Category 2: 63 – 67 dB (A) ▪ Category 3: 68 – 72 dB (A)

4.6 Assessment Benchmarks

The Assessment benchmarks for the application are prescribed by the category of assessment tables for the Rural Zone Code and the applicable overlays. **Table 9** identifies the following codes that are applicable to the assessment of the proposed development. Further, a comprehensive assessment of the proposed development against the primary codes is attached in **Appendix C** of this report.

In summary, this assessment coupled with the relevant supporting reports demonstrates that the proposed development is capable of fulfilling the Acceptable Outcomes or associated Performance Outcomes of each Code.

See **Table 9** for a summary of the assessment benchmark compliance.

Table 8 – Assessment Benchmarks

Primary Codes	Prescribed Secondary Codes	Overlay Codes
▪ Rural Zone Code	▪ Transport, Access and Parking Code ▪ Infrastructure Services Code ▪ Accommodation Activities Code	▪ Wetlands Overlay Code; ▪ Biodiversity Areas Overlay Code; ▪ Bushfire Hazard Overlay Code; ▪ Regional Infrastructure Corridor – Stock Route Overlay Code;

A comprehensive assessment of the relevant codes is included in **Appendix C** and in the relevant specialist reports (refer **Section 6**). These assessments demonstrate the proposal complies with the assessment benchmarks applicable to the development.

4.7 Local Government Infrastructure Plan

The Local Government Infrastructure Plan ('**LGIP**') integrates and coordinates land use planning and infrastructure planning and ensures the trunk infrastructure is planned and provided in an efficient and orderly manner.

The LGIP identifies the following in relation to the site:

- The site is not located within the Priority Infrastructure Area; and
- No future trunk infrastructure is mapped at or adjoining the site.

4.8 Approval Pathway

The proposed development is being progressed through a Development Application for a Material Change of Use under the Planning Act, with Western Downs Regional Council acting as the Assessment Manager.

As outlined in **Section 4.4**, the proposed Research and Technology Industry, Workforce Accommodation and High Impact Industry are subject to Impact Assessment under the Planning Scheme. Accordingly, the application will be assessed against the relevant provisions of the Planning Scheme, applicable State interests and referral agency requirements, and will be publicly notified in accordance with the Development Assessment Rules.

In parallel with the statutory development assessment process, the project proponent is pursuing declaration of the project as a Prescribed Project (e.g. a project of significant economic and social importance to Queensland or the region) under the *State Development and Public Works Organisation Act 1971* (the '**SDPWO Act**'). A prescribed project declaration, made by the Minister for State Development, Infrastructure and Planning (DSDIP), enables the Coordinator-General (CG) to work with local government, regulators and the proponent through the approvals process for the project, to ensure that there are no unreasonable delays.

Application processes, for example those made under the Planning Act, remain unchanged unless the CG issues a notice under SDPWO Act. For example:

- Progression notice – requires the decision-maker to progress the administrative processes necessary to complete the assessment process;
- Notice to decide – requires the decision-maker to make the relevant decision within a specified timeframe;
- Step-in notice – allows the CG (with the Minister's approval) to step in and assume responsibility for assessing and deciding on a project, in place of the decision-maker.

The proposed approval strategy has been developed to provide a clear and coordinated pathway for assessment while ensuring all relevant statutory requirements are appropriately addressed through the development assessment process.

5

Key Planning Matters

5 Key Planning Matters

The following section of this report provides further discussion of the key planning matters relevant to the proposed development. The assessment is intended to assist Council and the broader community in understanding the planning merits of the proposal and demonstrates how the development responds to the strategic intent and applicable provisions of the Planning Scheme.

Given the scale, complexity and significance of the proposed development, a number of planning matters warrant detailed consideration beyond the statutory planning framework outlined in **Section 3**. These matters have been informed by the accompanying technical assessments and provide the planning justification for the proposed development.

The key planning matters addressed in this section include:

- assessment against the key Performance Outcomes of the Western Downs Planning Scheme;
- land use compatibility and separation distances;
- the strategic merit of the proposed location;
- planning need and broader community benefit; and
- scenic amenity and visual impacts.

5.1 Assessment against Key Performance Outcomes

The proposed Research and Technology Industry, ancillary High Impact Industry (concrete batching plant) and Workforce Accommodation are generally consistent with the intent of the Western Downs Planning Scheme and achieve compliance with the majority of the applicable assessment benchmarks. This conclusion is supported by the suite of technical investigations prepared in support of the application, which demonstrate that the proposal can be developed without giving rise to unacceptable environmental, infrastructure or operational impacts.

Notwithstanding the above, two principal planning issues have been identified through the assessment process, namely:

- the proposed building height, having regard to PO1 of the Rural Zone Code;
- the proposal's amenity impacts, having regard to PO5 of the Rural Zone Code; and
- the appropriateness of the proposed land uses within the Rural Zone, having regard to the Strategic Framework and the purpose of the Rural Zone Code.

These matters are discussed below.

5.1.1 Building Height

The principal built form issue relates to PO1 of the Rural Zone Code, which seeks to maintain a low-rise built form having regard to existing landscape character values. While the proposed buildings are single storey, the primary Research and Technology Industry buildings have a maximum height of 13.23 metres to the roof ridge, exceeding the 10 metre building height prescribed by AO1 by approximately 3.23 metres. The building façades, however, are approximately 9.2 metres in height, with the additional height attributable to the pitched roof structure required to accommodate the specialised plant, equipment and internal clearances necessary for the operation of the facility.

The proposed building height is considered appropriate in the context of the development for several reasons. Firstly, the exceedance is relatively modest and is confined to the roof ridge rather than the building façade, reducing the perceived bulk and scale of the buildings. Secondly, the buildings are set back substantially from surrounding road frontages and property boundaries, with the primary data centre buildings located approximately 950 metres from Dalby–Kogan Road, significantly reducing their visual prominence.

Importantly, the subject land is identified within the Kogan East Special Industry Area, where the Planning Scheme anticipates large-scale industrial development requiring separation from sensitive land uses. In this context, the proposed building height reflects the functional requirements of a nationally significant Research and Technology Industry and does not undermine the underlying intent of PO1, having regard to the site's strategic location, generous setbacks and limited opportunity for adverse visual impacts.

5.1.2 Amenity (Acoustics)

The second key planning issue relates to PO5 of the Rural Zone Code, which requires development to avoid detracting from the amenity of the local area, having regard to matters including noise, traffic, lighting, visual amenity, odour and emissions.

The proposal is supported by a comprehensive suite of technical assessments, including specialist reports addressing lighting, air quality, traffic and acoustics. These assessments demonstrate that the proposal can be accommodated without unacceptable impacts in relation to all assessed amenity matters. The Noise Impact Assessment identifies that the current concept design does not achieve the adopted project noise criteria at a limited number of surrounding sensitive receivers and the proposed Workforce Accommodation. This outcome reflects the preliminary nature of the concept design, conservative assumptions adopted for plant and equipment, and the exceptionally low existing background noise environment surrounding the site.

Importantly, the Noise Impact Assessment concludes that noise does not represent a fundamental constraint to the development. Rather, it identifies a clear pathway for achieving compliance through the detailed design process, including refinement of equipment selection, acoustic treatments, building layout, shielding and operational controls. Verification through detailed acoustic modelling and commissioning will form part of the ongoing design development.

Having regard to the strategic designation of the site within the Kogan East Special Industry Area, the substantial separation distances to surrounding sensitive receivers, and the opportunity to refine the development through detailed design, it is considered that the proposal will not result in unacceptable amenity impacts. The proposal is therefore consistent with the underlying intent of PO5, with appropriate acoustic mitigation to be implemented as the project progresses through detailed design.

5.2 Land use compatibility

The compatibility of the proposed Research and Technology Industry, Workforce Accommodation and High Impact Industry (concrete batching plant) with the surrounding locality is a key planning consideration for the proposed development. It is acknowledged that the proposal is not wholly consistent with the purpose and intent of the Rural Zone and does not comply with Overall Outcomes 1, 4 and 6 of the Rural Zone Code. Collectively, these provisions seek to ensure the Rural Zone primarily accommodates rural land uses, that new enterprises are associated with rural production or natural resources, that rural landscape values are maintained through a low-rise built form, and that incompatible land uses do not encroach upon the rural environment. However, the Rural Zone Code also recognises that Special Industry uses requiring separation from sensitive land uses are appropriately located within identified Special Industrial Areas, and that development departing from the intent of the Rural Zone may be supported where overriding community need and sufficient planning justification are demonstrated. Accordingly, the appropriateness of the proposal is more appropriately assessed against the Strategic Framework.

The Strategic Framework identifies the subject land within the Kogan East Special Industry Area under Element 3.3.9 – Industrial Land Use and Development, which provides strategic direction for the location of future Special Industry uses.

“(6) The Special Industrial Areas identified on Strategic Plan Map 1 – Settlement Pattern indicates where future special industrial development is to be located. In this context, special industrial development includes high impact industry uses, renewable energy facility, battery storage facility and special industry as defined in Schedule 1 – Definitions.

(7) It is acknowledged that parts of the strategic plan suggest that Special Industry land uses (which include manufacturing and storage of explosives) are encouraged to occur within the identified Special Industrial Areas. These statements should not be limit the use of other Rural land for Special Industry land uses as more appropriate locations with the LGA may exist to accommodate the manufacturing and storage of explosives. The Special Industrial Area (Kogan East) show on Strategic Plan Map 1 – Settlement Pattern would not be supported for uses involving manufacturing and storage of explosives due to the adjacent Urban Areas zoning.”

Similarly, the Rural Zone code identifies the following Overall Outcome as being particularly relevant to the proposed development.

“(8) Special industry uses that require separation distances from sensitive land uses are supported and encouraged to locate in areas identified as Special Industrial Area identified on Strategic Plan Map 1 – Settlement Pattern”.

The proposal directly responds to these strategic planning directions. The Research and Technology Industry is a regionally significant industrial use that is appropriately located within an established energy and industrial precinct, immediately adjoining the Darling Downs Power Station, high-voltage electricity transmission infrastructure and other strategic infrastructure required to support its long-term operation. The proposal therefore represents the type of development envisaged for the Kogan East Special Industry Area and reinforces the strategic function of the precinct as a location for major industrial investment.

The associated Workforce Accommodation is integrated within the master planned development to support the specialised construction and operational workforce required for the establishment and ongoing operation of the data centre campus. The accommodation is directly linked to the primary industrial use and has been planned to function as part of the broader operational precinct rather than as a standalone residential development. Its location and design respond to the operational requirements of the project while ensuring appropriate separation from surrounding industrial and rural activities.

The proposal has been planned having regard to the existing and intended function of the locality. Key locational characteristics supporting the proposal include:

- the site's identification within the Kogan East Special Industry Area, which has been strategically identified for regionally significant industrial development;
- substantial separation distances from established urban areas and other sensitive land uses;
- direct access to major electricity generation and transmission infrastructure, supporting the operational requirements of the proposed Research and Technology Industry;
- proximity to the regional road network, enabling efficient access for construction, servicing and operational traffic; and
- sufficient land area to accommodate the proposed development while providing appropriate internal setbacks, operational separation and landscape buffers.

Relevant specialist reports have been prepared in support of the proposed development, addressing matters including ecology, traffic and transport, flooding, noise, air quality, servicing, visual amenity and other technical considerations. These assessments demonstrate that the proposed development can be appropriately accommodated on the site without resulting in unacceptable impacts on surrounding land uses or compromising the continued operation of existing industrial and agricultural activities. The relevant specialist reports are included within the appendices to this Planning Report.

5.3 Location and Strategic Merit

The subject site is strategically positioned within the Western Downs Region to accommodate the proposed Research and Technology Industry, Workforce Accommodation and High Impact Industry. Its designation within the Kogan East Special Industry Area, proximity to established energy infrastructure and substantial separation from urban communities collectively distinguish the site from more conventional rural land.

The site occupies a large landholding approximately 15 kilometres south-east of Kogan and is identified on Strategic Plan Map 1 – Settlement Pattern as part of the Kogan East Special Industry Area. This designation recognises the locality's long-term role in accommodating major industrial activities that require large sites, access to strategic infrastructure and separation from sensitive land uses.

The strategic merit of the site is supported by the following locational attributes:

- identification within the Kogan East Special Industry Area;
- proximity to major electricity generation and transmission infrastructure;
- access to Dalby–Kogan Road and the broader regional transport network;
- sufficient land area to accommodate staged development, internal infrastructure, security requirements and future expansion;
- commuting distance from established urban settlements;
- separation from established urban areas and sensitive land uses; and
- an established surrounding context of energy, industrial and rural activities.

The location of the site relative to the surrounding energy infrastructure and regional transport network is shown in **Figure 17** below.

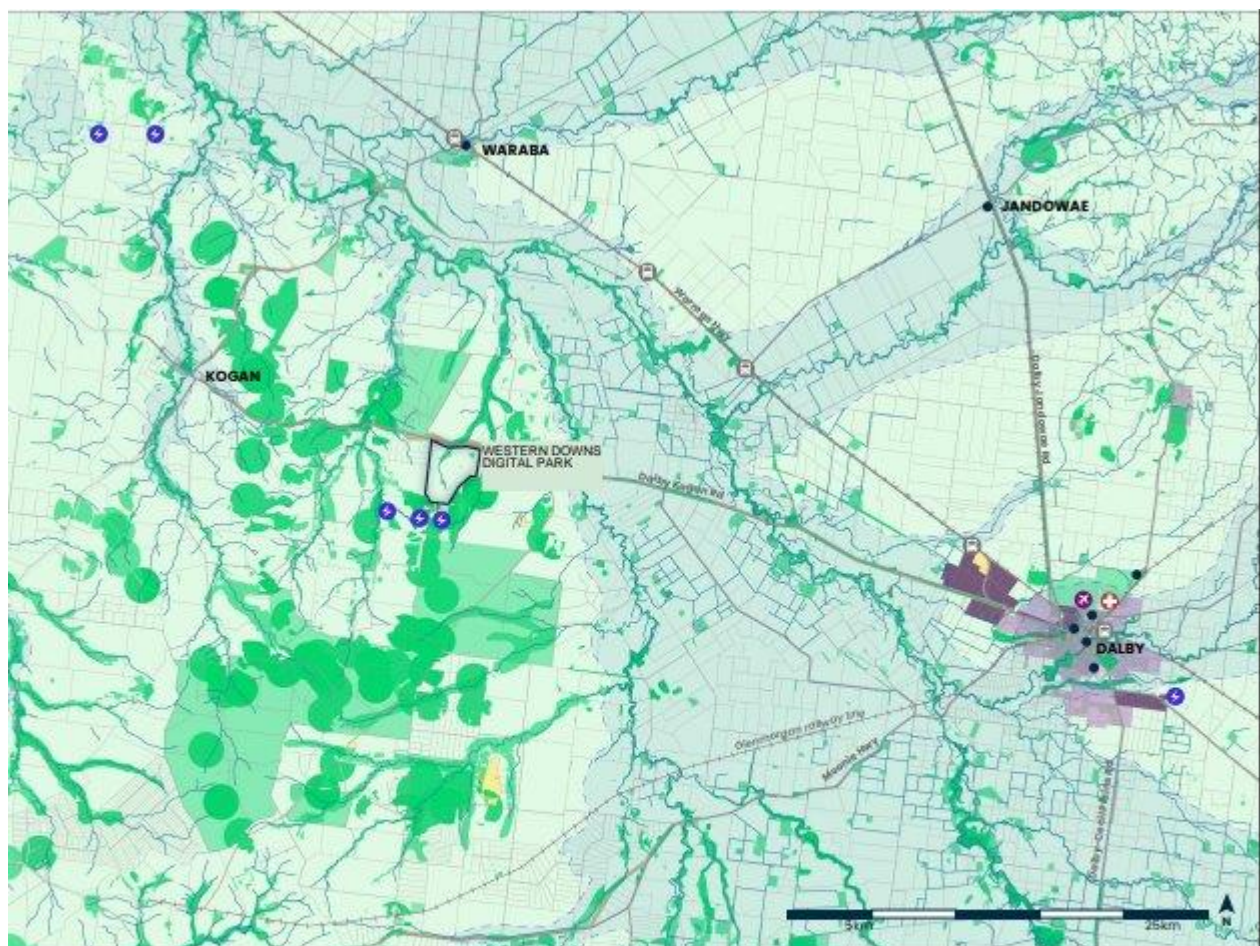


Figure 2 Local context map

LEGEND:	
■	Conservation and habitat
■	Flood-prone area
—	Watercourse
■	Housing
■	Industrial
■	Stock route
✕	Airport
✕	Hospital
✕	Power plant / solar farm
●	Educational facility

Figure

17 – Surrounding Context

Source: Urbis Ltd

The surrounding electricity network is a particularly important locational attribute. The site is positioned near the Braemar Substation, Braemar 1 and Braemar 2 Power Stations, Darling Downs Solar Farm and other established transmission and distribution infrastructure. An Ergon sub-transmission cable also traverses the site and connects with nearby distribution substations. The site is located within the Surat Basin North West Area Transmission Network, providing a strong infrastructure context for an energy-intensive use.

The locality also contains established gas infrastructure, including a pipeline along Dalby–Kogan Road and the eastern site boundary, together with a petroleum pipeline to the north-east. While these assets do not remove the need for detailed infrastructure planning and approvals, they reinforce the broader function of the area as an established energy and infrastructure precinct.

The size and configuration of the landholding provide flexibility to accommodate the proposed development in stages and to integrate access, security, utility infrastructure, environmental buffers and operational areas within a coordinated master plan. This allows the initial development footprint to be established without constraining future expansion or the continued operation of surrounding land uses.

The site is not currently connected to reticulated water or sewerage infrastructure. However, its scale provides the capacity to investigate staged and self-contained servicing solutions, including on-site water storage, alternative water supply and cooling options, and sustainable wastewater management. These matters are addressed through the detailed civil and utility servicing assessments supporting the application.

The Workforce Accommodation is located within the broader development area to support the construction, operation and maintenance requirements of the Research and Technology Industry. Its integration within the site reduces reliance on accommodation within surrounding communities and enables access, servicing and operational interfaces to be managed as part of the overall development.

Collectively, the site's strategic designation, proximity to major energy infrastructure, separation from sensitive land uses and ability to accommodate a staged industrial development establish a strong planning basis for the proposed development. These locational attributes provide a unique opportunity to deliver a major Research and Technology Industry within an established energy precinct while ensuring the development can operate efficiently and integrate with the surrounding land use pattern.

5.4 Planning Need and Community Benefit

The proposed development responds to a demonstrated need for additional digital infrastructure within Queensland and Australia while delivering significant economic and community benefits to the Western Downs Region. As a Research and Technology Industry supported by associated Workforce Accommodation and High Impact Industry, the proposal will facilitate investment in nationally significant digital infrastructure that supports Australia's growing reliance on cloud computing, artificial intelligence, advanced data processing and secure digital services.

Beyond responding to an identified infrastructure need, the proposal represents a substantial long-term investment in regional Queensland that will strengthen economic resilience, diversify the regional economy and support the continued growth of Queensland's digital economy.

5.4.1 Planning Need

Demand for digital infrastructure continues to increase as government, industry and businesses become increasingly reliant on cloud computing, artificial intelligence, advanced data processing and secure digital storage. These technologies require significant computing capacity supported by reliable energy infrastructure and large industrial sites capable of accommodating long-term expansion.

The proposed development responds to these emerging requirements by providing a purpose-designed Research and Technology Industry within an established industrial and energy precinct. The development will contribute to the continued evolution of the Western Downs Region beyond its traditional agricultural and energy sectors by supporting investment in advanced technology industries.

The proposal is also consistent with broader Queensland Government initiatives to strengthen the State's digital capability, encourage investment in emerging industries and support regional economic diversification. The project represents an opportunity to capitalise on existing infrastructure assets while facilitating long-term economic growth within a strategically identified industrial location.

5.4.2 Economic Contribution

The proposed development represents a significant long-term investment in the Western Downs Region and will deliver substantial economic benefits throughout both the construction and operational phases. Beyond the direct investment in the site, the proposal will support employment, strengthen regional supply chains and contribute to the continued diversification of the regional economy through the establishment of a nationally significant Research and Technology Industry.

Key economic outcomes include:

- Approximately **1,500 direct on-site construction jobs** over the four-year construction period.
- Approximately **1,400 ongoing direct on-site jobs** once the development reaches full operational capacity.
- An estimated **\$760 million in annual Gross Value Added (GVA)** to the Queensland economy during ongoing operation.

The economic contribution of the proposal extends beyond these measurable outcomes. The proposal will create opportunities for regional businesses through local procurement, contractor engagement and supporting industries, while fostering a highly skilled workforce across technical, professional and operational disciplines. The establishment of a major Research and Technology Industry within the Western Downs will also strengthen the region's economic resilience by broadening its industrial base and supporting continued investment in advanced industries.

Collectively, these outcomes demonstrate that the proposal will make a substantial and enduring contribution to the regional and State economies, extending well beyond the immediate benefits associated with the development itself. Further detail is provided in the Economic Benefits Assessment at **Appendix P**.

5.4.3 Community Benefit

In addition to its economic contribution, the proposed development will deliver a range of broader community benefits that support the long-term growth, resilience and diversification of the Western Downs Region.

The establishment of a nationally significant Research and Technology Industry will strengthen the region's role within Queensland's evolving digital economy, providing critical infrastructure that supports government, industry and businesses increasingly reliant on secure, high-capacity digital services. In doing so, the proposal has the potential to attract complementary industries and investment, reinforcing the Western Downs as an emerging location for advanced technology and innovation.

The development will also broaden employment pathways within the region by introducing highly skilled technical, engineering, operational and maintenance roles while expanding opportunities for workforce training and specialised skills development. These outcomes will assist in building a more diverse and resilient regional workforce, supporting long-term employment opportunities beyond the region's traditional agricultural and energy sectors.

More broadly, the proposal will support local businesses through procurement opportunities, contribute to the growth of the regional digital ecosystem and encourage complementary investment in infrastructure and industry. By leveraging the region's existing strategic energy infrastructure and industrial assets, the development will deliver enduring community benefits that extend beyond the project site and support the long-term prosperity of the Western Downs.¹

5.4.4 Housing Strategy

Considering the size of the data centre, there is likely a need to import workforce from beyond the region to deliver the construction works and support the ongoing operation and maintenance of the data centre facility, given the small local resident and worker population base.

While the construction workers will be accommodated within a worker's camp on-site, over the project delivery period, housing provision for future permanent workers will need to be considered. As such, a housing analysis and delivery strategy is being prepared to establish the scale of accommodation required, timing of delivery and the potential delivery options and pathways.

The strategy is being prepared in collaboration with Western Downs Regional Council, and will combine economic analysis, demographic and housing market research, land supply assessment and strategic

planning to provide a clear implementation pathway that supports workforce attraction, community outcomes and long-term regional growth.

In addition to a housing strategy the project team is preparing a report looking at the provision of Social Infrastructure (such as health and education facilities) that may be required with an influx of residents.

5.5 Scenic Amenity and Visual Character

The proposed data centre and associated development is significant in scale but is largely visually contained by established boundary vegetation, which effectively screens most on-site built elements from surrounding areas – particularly Dalby Kogan Road. Below is a response to Planning Scheme Policy 3 – Landscape Character Analysis Guidelines.

Site Condition: Largely cleared, agricultural or vacant land.

Visibility: Low (due to topography, existing buffers, and distance from sensitive receptors).

Response to Item 1: Visibility – Where can the development be seen from?

- **Assessment:** The proposed development has very limited visibility from significant viewpoints.
- **Significant Viewpoints:** The site is not visible from any designated Scenic Routes, National Parks, or elevated lookouts. Views are limited to Grahams Road, which is used primarily for access to existing energy infrastructure rather than tourism.
- **Sensitive Receptors:** There are no residential clusters or public parks within the primary viewshed.

Response to Item 2: Appearance – What will it look like when seen from significant viewpoints?

- **Assessment:** From the viewpoints available, the development will appear as low-profile industrial structures consistent with the functional requirements of the Special Industry strategic designation.
- **Scale and Form:** The built form is restricted to 12.3 metres in height, ensuring it has minimal impact on skyline when viewed from the road. The use of non-reflective, cladding will minimize visual "glare" and allow the structures to recede into the landscape.
- **Contextual Fit:** Because the site is already largely cleared, the development does not require the removal of significant stand-alone trees or remnant vegetation. It will appear as a consolidated footprint rather than a fragmented intrusion into the rural landscape.

Response to Item 3: Mitigation Measures

- **Assessment:** While visibility is low, the following measures are proposed to ensure the development does not have an unacceptable impact on the prevailing character of the area, or the wider scenic amenity of the Western Downs.
- **Vegetation Screening:** landscape buffer will be established along the boundary using native species (consistent with the Forested and Woodland Downs character). This will provide a "soft" edge to the industrial site.
- **Lighting Control:** All outdoor lighting will be shielded and directed downwards to prevent light spill into the night sky, preserving the rural dark-sky character.
- **Supporting Detail:** Refer to the Landscape Concept and Intent Plan for specific plant species, spacing, and proposed landscape integration.

As noted above, public views are principally limited to Grahams Road, which provides access to existing energy infrastructure (Darling Downs Power Station, Braemar 1 and Braemar 2 and Darling Downs Solar Farm). The existing infrastructure (including pylons, substations, access tracks, gas wells and ancillary facilities) already defines the scenic amenity and visual character and context of the locality, reducing the visual sensitivity of the site, and the visual impact of the proposed development.

Overall visual impact is moderate to low when considered across the landscape: localised visual change is observable from Grahams Road and nearby properties with direct lines of sight, but the combination of significant building setbacks and the wider industrial character lessens broader landscape-scale impacts.

Key landscape values in the area are largely associated with functional energy infrastructure rather than high scenic or conservation values; there are no notable scenic gateways or prominent public recreation vantage points within immediate view that would be substantially affected. The proposed development is expected to have a negligible impact on the landscape character of the region.

6 **Specialist Reporting**

6 Specialist Reporting

The proposed development has carefully considered specialist advice and been informed by numerous technical investigations. The following specialist reports have been prepared in support of the proposed development. The complete list of the specialist reports is included in the following appendices and summarised in the sections below.

- **Appendix C** – Western Downs Planning Scheme 2017 Benchmark Assessments;
- **Appendix D** – Architectural Drawings, prepared by *HDR*;
- **Appendix E** – Landscape Plans, prepared by *Urbis*;
- **Appendix F** – Traffic, Transport and Access Assessment, prepared by *MCG Consult*;
- **Appendix G** – Stormwater Management Plan, prepared by *MCG Consult*;
- **Appendix H** – Infrastructure Services Report, prepared by *MCG Consult*;
- **Appendix I** – Fire Safety Concept Strategy, prepared by *CORE Engineering Group*;
- **Appendix J** – Environmental Assessment Report, prepared by *ERM*;
- **Appendix K** – EPBC Self-Assessment Report, prepared by *ERM*;
- **Appendix L** – Air Quality Assessment, prepared by *ERM*;
- **Appendix M** – Noise Impact Assessment, prepared by *ERM*;
- **Appendix N** – External Lighting Assessment, prepared by *HDR*;
- **Appendix O** – Waste Management Plan, prepared by *Urbis*;
- **Appendix P** – Economic Benefit Assessment, prepared by *Urbis*;

A summary of the key findings of each of the technical reports is provided below.

6.1 Traffic, Transport and Access Assessment

A Traffic, Transport and Access Assessment has been prepared by *MCG Consult* and is provided at **Appendix F**. The assessment considers the transport implications of the proposed development, including site access arrangements, traffic generation, heavy vehicle movements, construction traffic and operational servicing requirements.

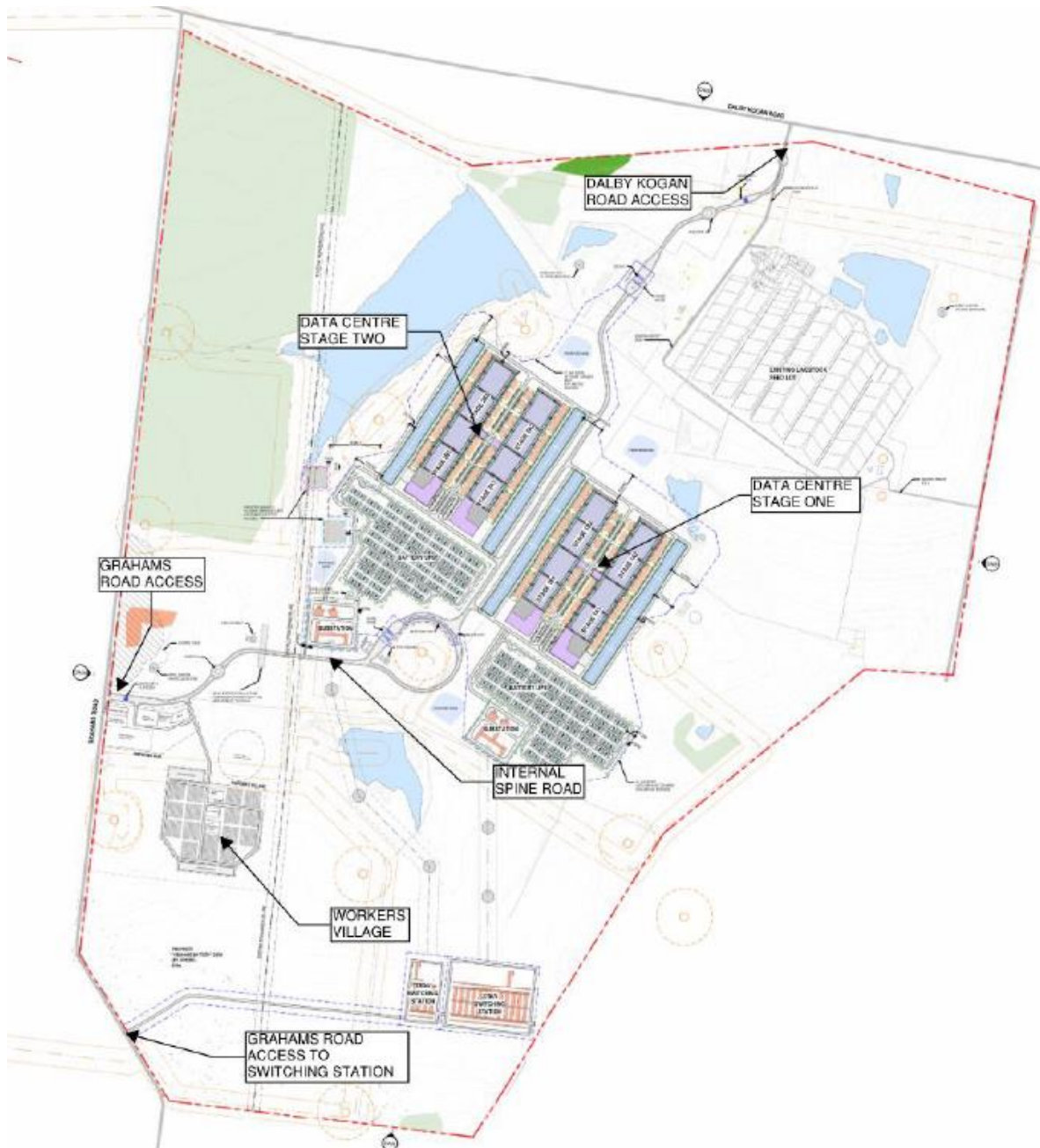
The assessment confirms that the proposed development can be appropriately accommodated by the surrounding road network, subject to the implementation of the recommended access upgrades, detailed design and ongoing consultation with the relevant road authorities. A summary of the key findings is provided below.

6.1.1 Site Access and Arrangements

The proposed development incorporates a comprehensive access strategy designed to safely and efficiently accommodate construction, operational and Workforce Accommodation traffic. As illustrated in **Figure 18**, the development is proposed to be accessed via two primary vehicular access points, comprising the principal access from Dalby-Kogan Road, generally utilising the existing feedlot access, and a secondary access from Grahams Road.

These access points are connected by an internal spine road, providing efficient circulation throughout the site, operational redundancy and separation between construction, operational and Workforce Accommodation traffic movements.

Figure 18 – Proposed Site Access Arrangements



Source: MCG Consult

In developing the proposed access strategy, the assessment has had regard to a range of site-specific considerations, including:

- available sight distances;
- surrounding road geometry;
- existing vegetation;
- flooding constraints;
- existing power and gas infrastructure; and
- the operational requirements of the proposed development.

The proposed internal road network has similarly been designed to accommodate the operational requirements of the development and includes:

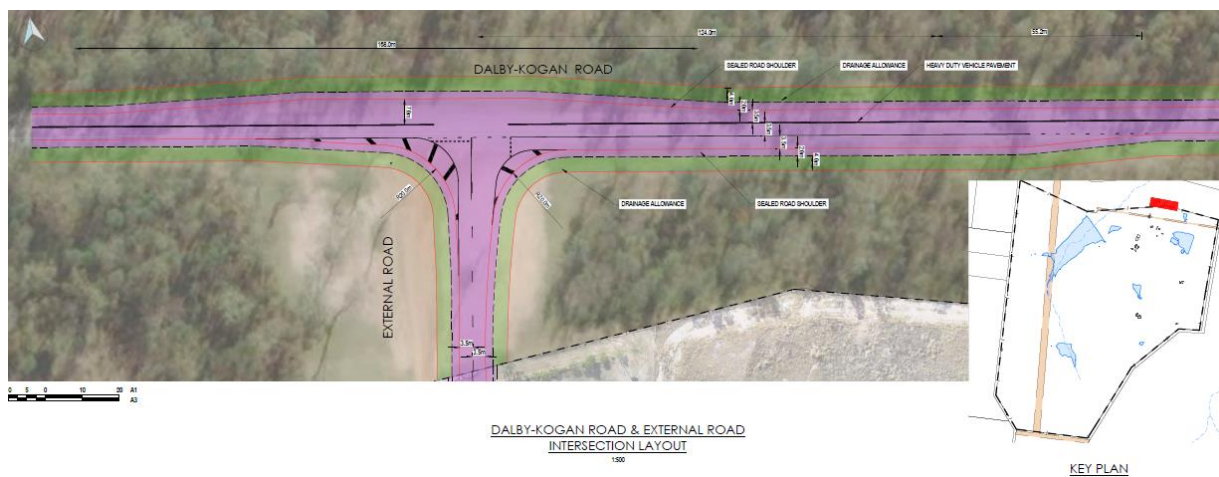
- an internal spine road connecting Dalby–Kogan Road and Grahams Road;
- dedicated internal circulation for construction, operational and Workers' Village traffic;
- heavy vehicle turning and manoeuvring areas;
- security gate queue storage and vehicle rejection areas;
- bus access within the Workers' Village; and
- appropriate internal traffic calming and speed management measures.

To facilitate safe access to the site, the Transport Impact Assessment identifies upgrades to the surrounding road network and intersection arrangements.

Dalby–Kogan Road Upgrade

The principal access from Dalby–Kogan Road is proposed to incorporate an Auxiliary Left Turn (Short) (AUL(s)) treatment and a Basic Right Turn (BAR) treatment (see **Figure 19** below). These upgrades have been assessed against the relevant Austroads warrants and are considered suitable to accommodate the anticipated construction and operational traffic volumes.

Figure 19 – Dalby–Kogan Road Access Upgrade

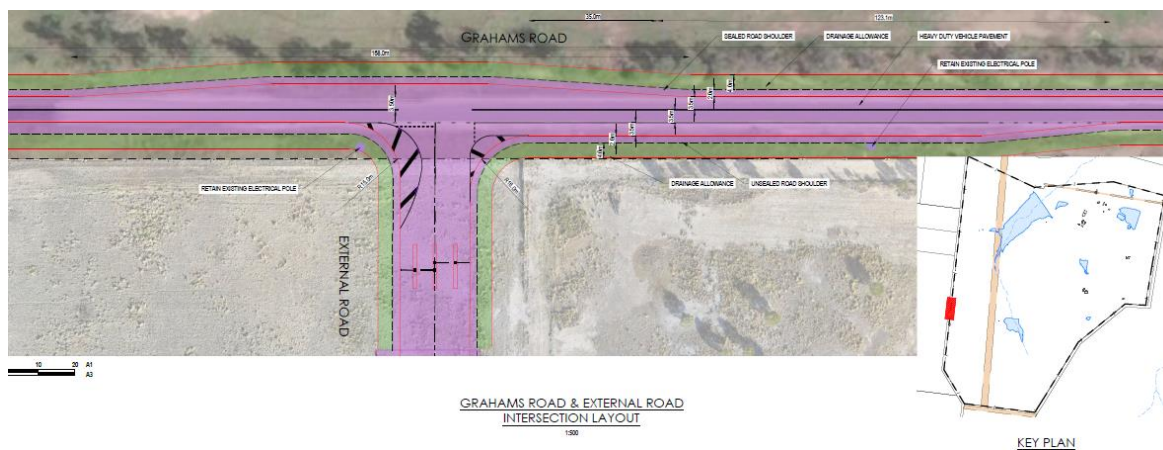


Source: MCG Consult

Grahams Road Upgrade

The secondary access from Grahams Road is proposed to operate as a priority-controlled rural intersection incorporating Basic Left Turn (BAL) and Basic Right Turn (BAR) treatments (see **Figure 20** below).

Figure 20 – Grahams Road Access Upgrade



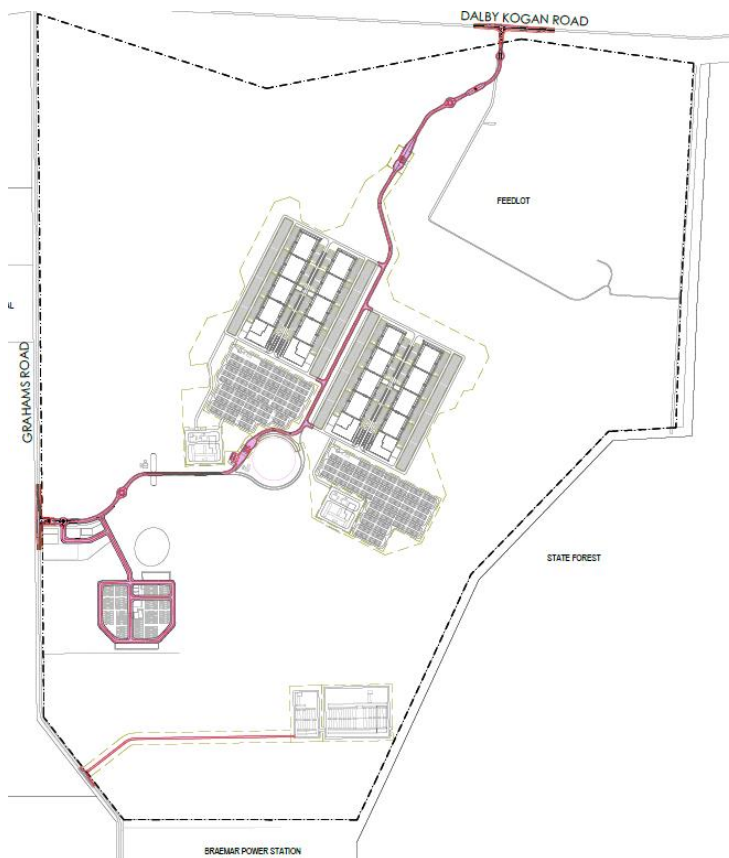
Source: MCG Consult

Internal Road Layout

Separate access arrangements are also proposed for the Workforce Accommodation from Grahams Road. These accesses are intended to operate as priority-controlled rural intersections, with auxiliary left-turn and basic right-turn treatments provided where warranted by future traffic volumes.

Beyond the external access arrangements, the proposed development incorporates an extensive internal road network centred around a spine road connecting Dalby-Kogan Road and Grahams Road. As illustrated in **Figure 21**, the network has been designed to provide efficient access throughout the site while accommodating construction traffic, heavy vehicles, buses servicing the Workforce Accommodation, emergency vehicles and day-to-day operational movements. The internal layout also incorporates security gate queue storage, vehicle rejection areas, turning facilities and traffic calming measures to support the safe and efficient operation of the development.

Figure 21 - Indicative Internal Road Network



Source: MCG Consult

6.1.2 Traffic Generation and Vehicle Movement

The assessment considers both construction and operational traffic associated with the ultimate master planned development, including the Workforce Accommodation.

Operational traffic generation is expected to remain relatively low due to the Workforce Accommodation accommodating the majority of the operational workforce and significantly reducing external commuter movements through a predominantly bus-in/bus-out workforce model. The assessment estimates that the ultimate development will generate approximately:

- 86 light vehicle movements per day;
- 4 bus movements per day;
- 30 heavy vehicle movements per day; and
- approximately 120 total vehicle movements per day.

Based on these assumptions, operational peak hour traffic generation is estimated at approximately **24 vehicles per hour**. Traffic modelling demonstrates that these additional movements remain well within the available capacity of the surrounding road network, with Dalby–Kogan Road, Grahams Road and the Warrego Highway all continuing to operate efficiently under forecast operational conditions.

6.1.3 Heavy Vehicle Routes and Road Impacts

Heavy vehicle movements associated with both the construction and operational phases have been assessed as part of the Traffic, Transport and Access Assessment.

Dalby–Kogan Road has been identified as the principal heavy vehicle access route, forming part of the gazetted B-Double network and PBS Level 3A network, permitting Type 1 Road Train access. Grahams Road is not currently gazetted for B-Double or Higher Mass Limit (HML) vehicles.

The assessment identifies no significant heavy vehicle access constraints associated with the proposed development. Matters including potential gazettal of Grahams Road for B-Double access, extension of HML access, Oversize/Overmass (OSOM) approvals, confirmation of route clearances and flood contingency access arrangements will be addressed through detailed design and consultation with the relevant road authorities.

6.1.4 Preliminary Construction Traffic Management

The assessment identifies the construction phase as the period generating the highest traffic demand, particularly during the establishment and early construction phases when bulk earthworks, material deliveries and major construction activities occur concurrently.

At peak construction, the assessment estimates approximately:

- 200 light vehicle movements per day;
- 230 heavy vehicle movements per day (including concrete batching plant deliveries); and
- approximately 430 total construction vehicle movements per day.

T The assessment also considers the combined scenario of **Stage 1 operations occurring concurrently with Stage 2 construction** and concludes that the surrounding road network is capable of accommodating the forecast traffic demand.

Construction traffic management will continue to be refined through detailed design, including preparation of a Construction Traffic Management Plan and ongoing consultation with the relevant road authorities regarding heavy vehicle operations, haulage routes and access arrangements.

6.1.5 Operational Servicing and Access

T Operational servicing requirements are expected to be relatively modest compared to the construction phase and will primarily comprise diesel generator refuelling, water deliveries, waste collection, general deliveries, maintenance activities and occasional mobile crane access.

The assessment estimates that the ultimate development will generate approximately **30 heavy vehicle movements per day**, with fewer than **10 heavy vehicle movements during any peak hour**. These servicing requirements can be accommodated within the proposed internal road network, which has been designed to facilitate heavy vehicle manoeuvring, loading activities and security screening while maintaining appropriate separation between operational traffic, servicing vehicles and Workforce Accommodation movements.

Overall, the assessment confirms that the proposed access strategy, traffic generation and servicing arrangements are appropriate for the scale and operational requirements of the proposed development, subject to detailed design and implementation of the recommended access treatments.

Further detail is provided in the Traffic, Transport and Access Assessment at **Appendix F**.

6.2 Stormwater Management Plan

A Stormwater Management Plan (SWMP) has been prepared by *MCG Consult* and is provided at **Appendix G**. The assessment considers the proposed stormwater management strategy for the proposal, including stormwater quantity and quality management, erosion and sediment control, and the long-term operation of the stormwater network.

The assessment confirms that the proposed development can be serviced by an integrated stormwater management system that appropriately manages stormwater runoff while maintaining the existing downstream drainage regime. A summary of the key findings is provided below.

6.2.1 Stormwater Management Strategy

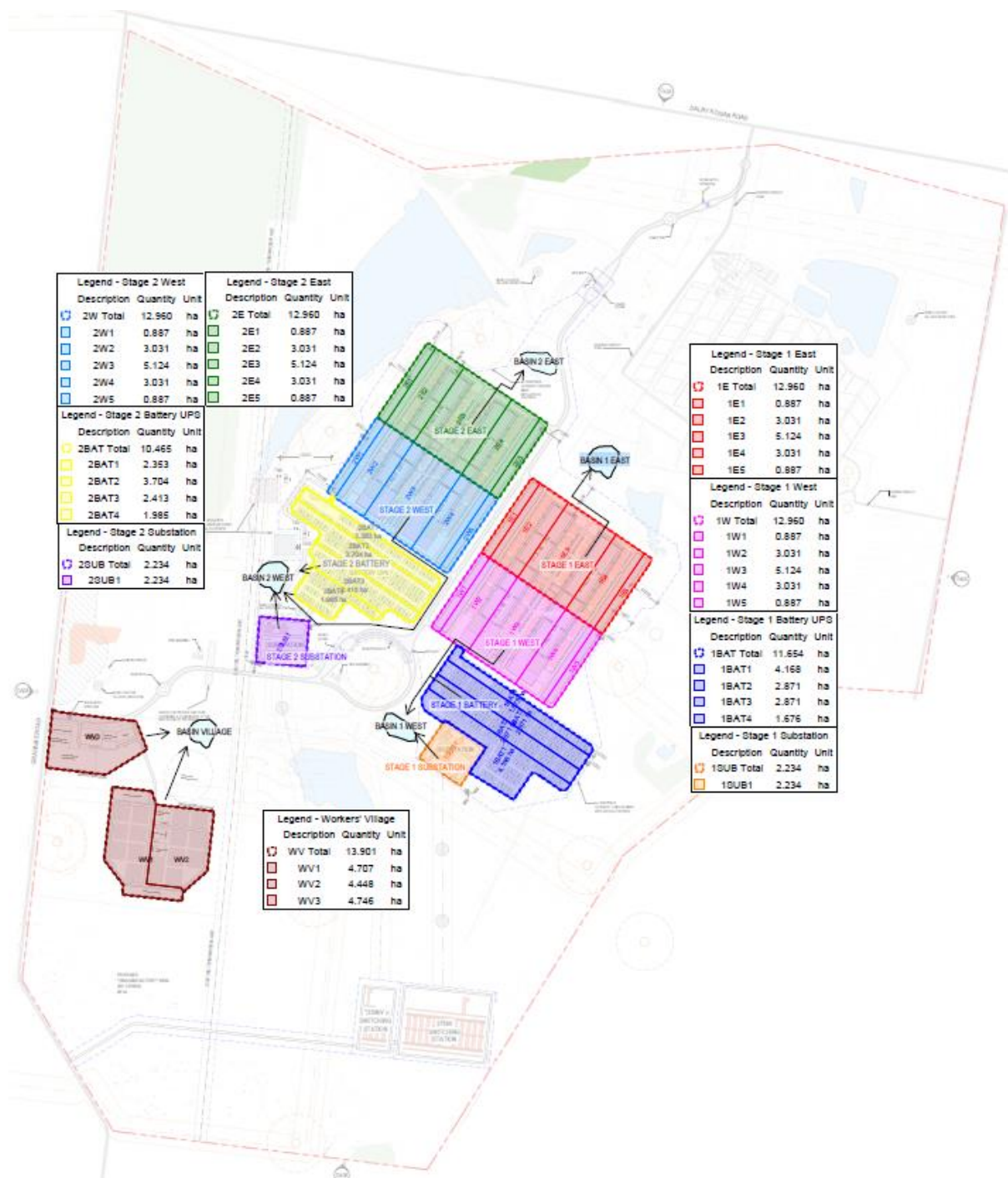
The site generally drains from south to north through a network of existing drainage lines and farm dams before discharging beneath Dalby–Kogan Road into the downstream creek system. The proposed stormwater management strategy retains this established drainage regime and lawful point of discharge while incorporating an integrated network of drainage infrastructure and detention basins to manage runoff generated by the development.

Developed in accordance with the Queensland Urban Drainage Manual (QUDM), the strategy has been informed by hydrological modelling to determine the stormwater attenuation requirements associated with the proposed development. Stormwater runoff generated across the site will be collected through the internal drainage network before being conveyed to a series of detention basins (see **Figure 22**) strategically located to service the staged development catchments, comprising:

1. **Basin Village**, servicing the Workers Village across three separate catchments;
2. **Basin 1 West**, servicing the Stage 1 West, Stage 1 Battery and Stage 1 Substation catchments;
3. **Basin 1 East**, servicing the Stage 1 East catchment;
4. **Basin 2 West**, servicing the Stage 2 West, Stage 2 Battery and Stage 2 Substation catchments; and
5. **Basin 2 East**, servicing the Stage 2 East catchment.

The assessment confirms that the proposed stormwater management system has been designed to maintain pre-development peak discharge rates and the existing lawful point of discharge, thereby achieving a non-worsening outcome for downstream flooding, scour and erosion.

Figure 22 – Post-Development Catchment Plan



Source: MCG Consult

6.2.2 Water Quality and Operational Management

In addition to managing stormwater quantity, the proposed development incorporates a range of Water Sensitive Urban Design (WSUD) measures, including sedimentation basins, bioretention basins, vegetated swales and rainwater harvesting infrastructure. These measures will treat stormwater prior to discharge and have been demonstrated through the supporting modelling to achieve the adopted stormwater quality objectives.

Roof runoff will also be captured for non-potable reuse throughout the development, with approximately **60kL of rainwater storage proposed for each development stage**, comprising 30kL for the eastern

catchment and 30kL for the western catchment. This will assist in reducing both potable water demand and the volume of stormwater discharged from the site.

Construction-phase erosion and sediment controls will be implemented in accordance with the *International Erosion Control Association (IECA) Best Practice Erosion and Sediment Control Guidelines*. A detailed Erosion and Sediment Control Plan will be prepared during detailed design and will address matters such as staged clearing, the diversion of clean runoff, temporary sediment controls and progressive stabilisation of disturbed areas.

Ongoing inspection and maintenance requirements have also been identified for the detention basins, bioretention systems, vegetated swales and rainwater harvesting infrastructure. These measures will support the continued operation of the stormwater system in accordance with its design intent. Further information is provided at **Appendix G**.

6.3 Infrastructure Services Report

HDR has prepared an Infrastructure Services Report (**Appendix H**) to assess the servicing requirements of the proposed development and demonstrate that adequate infrastructure can be provided throughout both the construction and operational phases. The report considers the provision of electricity, telecommunications, water supply, fire water infrastructure, wastewater servicing and gas infrastructure, having regard to the site's rural location and surrounding utility networks.

6.3.1 Electrical Infrastructure

The report identifies the subject site as being strategically located within one of Queensland's most significant energy precincts, immediately adjoining the Braemar 330kV Substation and in close proximity to the Braemar 1 and 2 Gas Power Stations, Darling Downs Power Station and multiple 275kV and 330kV transmission lines. Existing high-voltage transmission infrastructure is located approximately 180–400 metres south of the site, approximately 600 metres from the Braemar Transmission Substation and includes an existing 275kV transmission line traversing the western portion of the site.

The electrical high voltage design for Western Downs Digital Park aims to provide adequate capacity and reliability for the maximum demand required of four data buildings, each with a total electrical demand of 540 MVA. Across all four buildings, the campus reaches approximately 2,160 MVA of total facility demand.

The high voltage system is designed to meet the maximum demand while maintaining N+1 redundancy continuously from the switching stations downwards across the site. A dual power conversion and battery energy storage system is integrated between each substation and the data buildings, providing two hours of autonomy per stage, decoupling the load from the network, and allowing the storage to participate in network services

6.3.2 Telecommunications Infrastructure

The Infrastructure Services Report confirms that telecommunications services can be provided through a resilient fibre network supported by multiple geographically diverse connection paths. Initial investigations identified existing Telstra telecommunications infrastructure adjoining the site, with the preferred servicing strategy providing four independent fibre entry points to maximise redundancy and operational resilience. Internal fibre pathways are also proposed to maintain a minimum separation of 20 metres between diverse routes throughout the campus.

The preferred telecommunications strategy includes dedicated interconnection points to Telstra, Powerlink/QCN, the Dalby Exchange and Vocus/NextGen networks, providing multiple carrier options and future scalability appropriate for a development of this nature. Collectively, these arrangements ensure a reliable communications network capable of supporting the long-term operational requirements of the proposed data centre campus.

6.3.3 Water Infrastructure

The report confirms there is **no reticulated town water supply within approximately 15 kilometres** of the site. Consequently, the development will be serviced through an independent on-site water supply designed to accommodate both construction and operational demands. Water demand is primarily associated with domestic potable and non-potable uses together with one-off commissioning of the cooling systems, with no ongoing process water required for cooling during normal operations.

Peak water demand is anticipated during construction when the temporary workers' village is occupied, with demand estimated at approximately 322kL/day (270kL/day potable and 52kL/day non-potable), together with an additional non-potable demand of approximately 200kL/day for dust suppression and construction activities. By comparison, operational demand for a single completed data building reduces to approximately 16.5kL/day, reflecting the significant reduction in workforce following construction. In addition, each building will require approximately 26.9ML of water for one-off commissioning of the closed-loop cooling systems.

The preferred servicing hierarchy prioritises processed Coal Seam Gas (CSG) water during construction, followed by rainwater harvesting during operation, supported by modular potable water treatment, staged water storage, membrane bioreactor (MBR) wastewater treatment and reuse of recycled water for irrigation and other non-potable demands. The report concludes that these measures provide a practical and sustainable long-term water servicing solution for the development.

6.3.4 Fire Water Infrastructure

Given the absence of reticulated town water infrastructure, the report proposes a dedicated on-site fire water system to service both the data centre campus and temporary workers' accommodation. The primary data centre will be serviced by **two 700,000-litre fire water storage tanks**, together with two duty/standby fire pumps, a combined hydrant and sprinkler ring main, dedicated booster assemblies and an additional two 700,000-litre backup storage tanks to replenish the system following any emergency event.

A separate fire hydrant system is proposed for the workers' accommodation comprising two 144,000-litre storage tanks, dedicated duty and standby fire pumps, reticulated hydrant infrastructure and backup storage. Collectively, these measures provide an appropriate level of fire protection consistent with the scale and operational requirements of the proposed development.

6.3.5 Wastewater Servicing

As with water supply, there is no reticulated sewerage network servicing the site. Accordingly, wastewater generated by the development will initially be managed through on-site holding tanks and removal by licensed contractors during construction where required. Longer-term servicing will be provided through an on-site wastewater treatment facility capable of treating wastewater generated by both the operational campus and temporary workers' accommodation.

The preferred long-term servicing strategy incorporates an on-site sewage treatment plant with treated effluent reused for toilet flushing, washdown and irrigation, together with dedicated land application areas for disposal where required. The report confirms that sufficient land has been identified within the site to accommodate the treatment plant and associated irrigation areas, with the detailed design to be progressed during subsequent project stages.

6.3.6 Gas Infrastructure

The report identifies extensive existing gas infrastructure both within and adjoining the subject site. Existing QGC gas gathering pipelines, water gathering pipelines and drilled gas wells traverse the landholding, while a high-pressure APA gas transmission pipeline extends along the northern boundary. Additional gas transmission infrastructure operated by Verbrec-BPGL is located along the eastern and southern boundaries, with further Arrow Energy gas infrastructure also identified adjacent to the site.

The Infrastructure Services Report confirms these existing utility corridors have been incorporated into the proposed master plan and will continue to be managed through detailed design and ongoing consultation with the relevant asset owners. The presence of this existing infrastructure reinforces the strategic location of the site within an established energy precinct while ensuring the continued protection and operation of existing utility assets.

6.4 Fire Safety Concept Strategy

A Fire Safety Concept Strategy has been prepared by *CORE Engineering Group* and is provided at **Appendix I**. The strategy establishes the preliminary fire safety approach for the proposed development, including the management of fuels, batteries and electrical infrastructure associated with its operation.

The strategy has been developed with regard to the National Construction Code (NCC) 2022, relevant Australian Standards and Queensland requirements for the storage and handling of dangerous goods. It will continue to inform the building design, fire protection systems and emergency response arrangements as the project progresses through detailed design.

6.4.1 Hazardous Materials & Battery Systems

The principal hazardous materials identified within the development comprise:

- approximately **186,336L of diesel** stored within generator belly tanks, classified as Class C1 Dangerous Goods;
- lithium-ion batteries contained within **Integrated Power Units (IPUs)**, classified as Class 9 Dangerous Goods; and
- approximately **3,756,000L of transformer oil** within the High Voltage Switching Station (HVSS), noting that transformer oil is not classified as a Dangerous Good under the *Work Health and Safety Regulation 2011*.

Lithium-ion batteries associated with the IPUs will be located externally to the data halls within dedicated two-hour fire-rated compartments. This allows the battery systems to be physically separated from the principal operational areas and managed through dedicated fire protection and monitoring measures.

These measures include enhanced fire suppression, off-gas detection, mechanical ventilation and smoke detection. Together, they are intended to provide early warning of a battery fault, limit the potential for fire spread and support safe intervention by emergency responders.

Separate battery-backed power conditioning systems compounds (which are required for the operational of the development as an ancillary component) are also proposed as part of the development. These compounds will be treated as standalone Class 10 structures. The preliminary strategy provides for perimeter access, separation between battery banks, hydrant coverage and the application of recognised battery safety standards.

The assessment notes that no lithium-ion batteries are proposed within the data halls themselves. Rather, battery storage is confined to dedicated IPUs located external to the buildings. This arrangement significantly reduces the potential fire hazard within the operational data halls and allows battery risks to be managed through dedicated fire protection and monitoring systems.

6.4.2 Fire Protection, Access and Containment

The fire protection strategy will combine automatic detection and suppression systems with appropriate hydrant coverage, emergency vehicle access and separation between key infrastructure. The detailed design of these systems will be coordinated with the Queensland Fire Department (QFD) and relevant building certifier as the building and operational layouts are further developed.

Particular consideration has been given to the containment of potentially contaminated firewater. Runoff generated through sprinkler activation or fire brigade intervention within the IPUs will be directed to a

dedicated firewater containment tank with an approximate capacity of **500,000L**. Associated drainage controls and emergency procedures will prevent contaminated runoff from leaving the site and allow it to be appropriately contained and managed following an incident.

The storage and handling of diesel and other hazardous materials will also be subject to the relevant containment, separation and operational requirements under Queensland dangerous goods legislation. Further fire engineering and hazard studies will be undertaken where required to support detailed design and subsequent regulatory approvals.

Further information on the proposed systems, performance requirements and design standards is provided in the Fire Safety Concept Strategy at **Appendix I**.

6.5 Environmental Assessment Report

An Environmental Assessment Report has been prepared by *Environmental Resources Management Australia Pty Ltd (ERM)* and is included at **Appendix J**. The assessment examines the ecological values of the site, including vegetation communities, waterways, habitat for protected flora and fauna, Matters of State Environmental Significance (MSES) and Matters of Local Environmental Significance (MLES).

Field investigations confirm that the site is predominantly characterised by modified agricultural land, with ecological values generally confined to isolated areas of remnant vegetation, waterways and associated habitat features. This existing land condition has provided flexibility to position the proposed development within areas of comparatively lower ecological value while avoiding or minimising impacts to more sensitive environmental features where practicable.

6.5.1 Existing Ecological Values

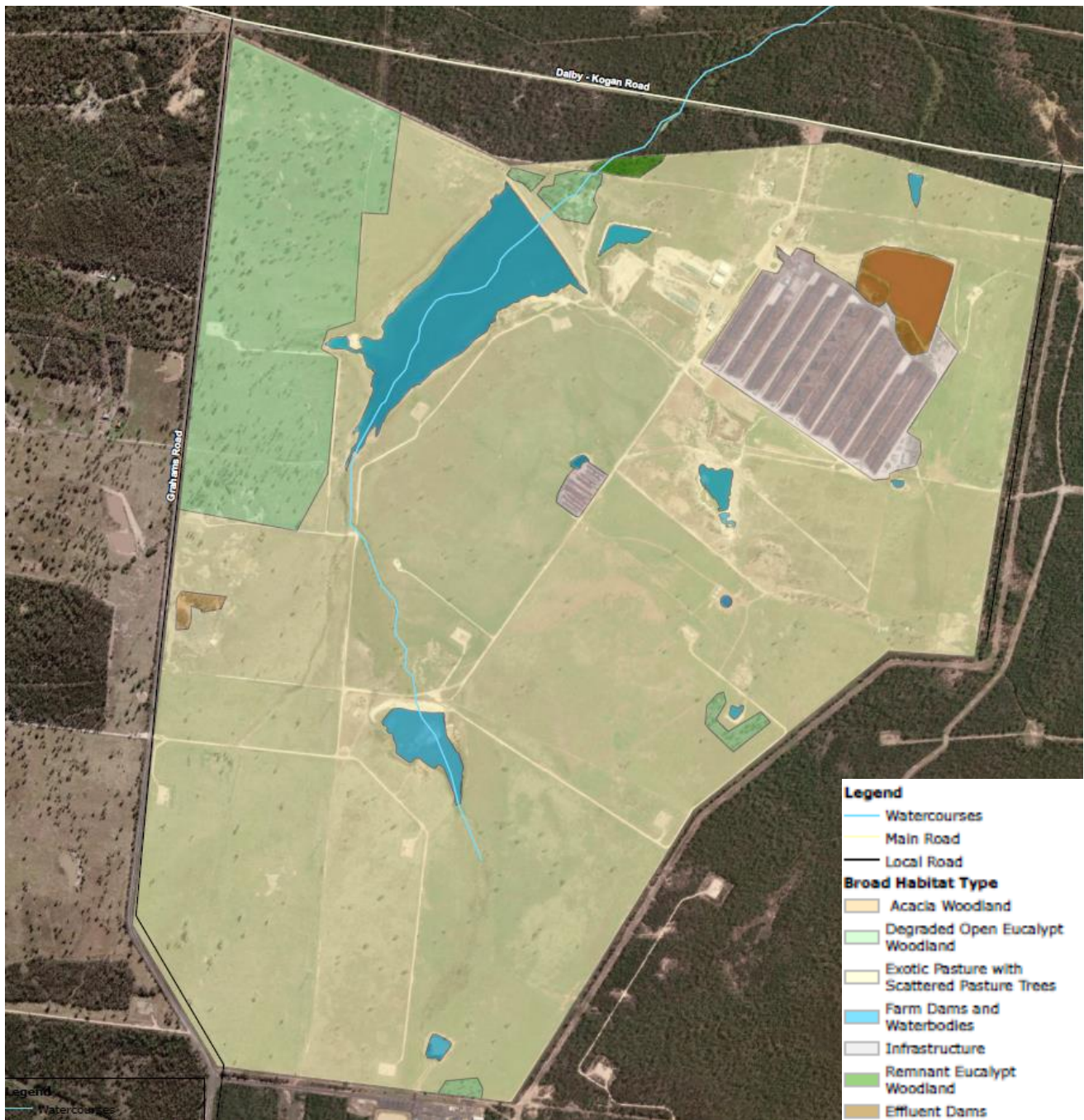
The site is dominated by **Category X regulated vegetation**, comprising exotic pasture with scattered trees across the majority of the Project Area. Smaller pockets of degraded open eucalypt woodland, fragmented acacia woodland, farm dams, waterbodies and a limited area of remnant eucalypt woodland occur throughout the site.

Five broad habitat types were identified across the project site:

- exotic pasture with scattered trees (the predominant habitat type);
- degraded open eucalypt woodland;
- fragmented acacia woodland;
- farm dams and waterbodies with varying levels of fringing vegetation; and
- a small area of remnant eucalypt woodland near the northern boundary.

These habitat types are illustrated in **Figure 23**.

Figure 23 – Habitat Types within the Subject Site



Source: ERM

6.5.2 Matters of State & Local Environmental Significance

Ecological investigations identified a relatively limited extent of MSES within the project site. These comprise approximately **1ha of Category B remnant vegetation**, vegetation associated with a mapped watercourse, known habitat for the Kogan waxflower, and potential habitat for several species protected under the *Nature Conservation Act 1992* (the '**NC Act**'). No state conservation areas, essential habitat, high ecological value wetlands, declared fish habitat areas or legally secured offset areas were identified within the site.

The **Kogan waxflower** is listed as Endangered under the NC Act and was recorded along the western boundary of the site, adjoining Grahams Road. ERM has identified approximately 5.1ha of supporting habitat, based on a 100-metre buffer around the observed individuals.

This ecological value is most relevant to any works proposed along Grahams Road. Where future works require the removal of an identified plant or clearing within the supporting habitat, a **Protected Plant Clearing Permit** may be required before clearing commences. The final extent of any potential impact will be confirmed through detailed design of the road and access works.

The planning scheme also identifies three existing waterbodies within the site as wetlands of **High Ecological Significance**. The proposed development has been planned with regard to these mapped features, including the application of appropriate setbacks and consideration of their associated ecological and hydrological functions.

6.5.3 Ecological Response

The proposed disturbance footprint has been developed to respond to the site's ecological constraints by directing development toward the predominantly cleared and fragmented portions of the landholding. This approach reduces interaction with remnant vegetation, waterways and identified habitat features while allowing further opportunities to avoid and minimise impacts during detailed design.

Consistent with this approach, ERM concluded that the proposal is **not expected to result in a Significant Residual Impact** on any identified MSES. This conclusion reflects the highly modified condition of the site, the limited extent of remnant ecological values and the ability to incorporate appropriate setbacks, mitigation measures and design refinements as the project progresses.

The ecological recommendations identified by ERM will be carried forward into the detailed design, construction and operational phases of the development, with the supporting report providing the detailed mapping, species assessments and Significant Residual Impact assessment. For further detail, refer to **Appendix J**.

6.6 EPBC Act Self-Assessment Report

ERM has prepared an EPBC Act Self-Assessment Report to determine whether the proposed development is likely to result in a significant impact on any Matters of National Environmental Significance (MNES) protected under the *Environment Protection and Biodiversity Conservation Act 1999* (the '**EPBC Act**'). The assessment, included at **Appendix K**, draws upon desktop investigations and ecological field surveys completed in June 2026 and has been prepared in accordance with the *Significant Impact Guidelines 1.1 – Matters of National Environmental Significance*.

6.6.1 Matters of National Environmental Significance

The assessment identified ten (10) threatened and migratory species listed under the EPBC Act with the potential to occur within the broader locality based on habitat availability and known species distributions. These include:

- Koala;
- Diamond Firetail;
- Australian Painted Snipe;
- Curlew Sandpiper;
- Latham's Snipe;
- Sharp-tailed Sandpiper;
- White-throated Needletail;
- Fork-tailed Swift;
- Common Sandpiper; and
- Pectoral Sandpiper.

Despite these potential occurrences, field investigations did not identify any listed threatened ecological communities or record any EPBC Act listed threatened or migratory species within the project site. The assessment notes that the site has been substantially modified through historical agricultural activities and provides relatively limited habitat when compared with the surrounding landscape.

6.6.2 Significant Impact Assessment

ERM assessed the proposal against the relevant significant impact criteria for each species considered likely or potentially present. The assessment concludes that the proposed development is **unlikely to result in a significant impact on any MSES**, having regard to:

- the absence of threatened ecological communities within the site;
- no recorded EPBC Act listed species during field investigations;
- the limited extent and generally low ecological value of potential habitat;
- the absence of important populations or critical habitat; and
- the availability of more suitable habitat within the surrounding landscape.

Accordingly, ERM concludes that the proposed development **does not require referral** to the Australian Government under the EPBC Act. The detailed –assessment, including the significant impact assessments for each potentially occurring species, is provided in **Appendix K**.

6.7 Air Quality Assessment

An Air Quality Assessment has been prepared by *ERM* and is included at **Appendix L**. The assessment considers potential air quality impacts associated with both the construction and operation of the proposed development, including emissions from temporary construction activities, backup diesel generators and their potential effects on nearby sensitive receptors, including the proposed Workforce Accommodation.

6.7.1 Construction Air Quality

During construction, the principal air quality impacts are expected to arise from earthworks, vehicle movements, material handling and temporary concrete batching activities. These emissions are anticipated to be localised and temporary in nature and are consistent with those typically associated with major construction projects.

The assessment identifies a range of established mitigation measures to minimise dust generation, including progressive stabilisation of disturbed areas, watering of exposed surfaces, management of vehicle movements and speeds, appropriate handling of fine materials, and covering loads during transport. Dust associated with the temporary concrete batching plant can likewise be managed through standard industry practices, including enclosed material transfers, filtration systems and stockpile management.

The assessment also considers the decommissioning of the existing feedlot effluent ponds, noting that short-term odour emissions may occur during these works. These impacts are expected to be temporary and can be managed through staged removal, minimising the exposure of residual material and implementing appropriate handling and disposal practices.

6.7.2 Operational Air Quality

Operational air quality modelling focused primarily on emissions from the proposed backup diesel generators. Under routine operating conditions, the assessment predicts compliance with the relevant air quality objectives at surrounding sensitive receptors, including the proposed Workforce Accommodation. Where cumulative particulate matter concentrations are elevated, the assessment concludes that the contribution from the proposed development is negligible and largely reflects existing background conditions.

A separate assessment of a conservative emergency operating scenario identified a short-term exceedance of the nitrogen dioxide (NO₂) criterion at the nearest off-site sensitive receptor and across parts of the proposed Workforce Accommodation. ERM notes that this scenario represents an unlikely operating condition based on conservative modelling assumptions and is therefore expected to overstate potential operational impacts.

6.7.3 Design Refinement

The assessment identifies a pathway to addressing the predicted short-term NO₂ impacts through further design refinement. Opportunities identified include selecting generators with improved emissions performance, increasing exhaust stack heights, refining generator locations and configurations, and incorporating appropriate operational controls.

The final generator specifications and exhaust arrangements will be confirmed during detailed design and supported by updated air quality modelling to demonstrate compliance with the applicable assessment criteria before the facility becomes operational.

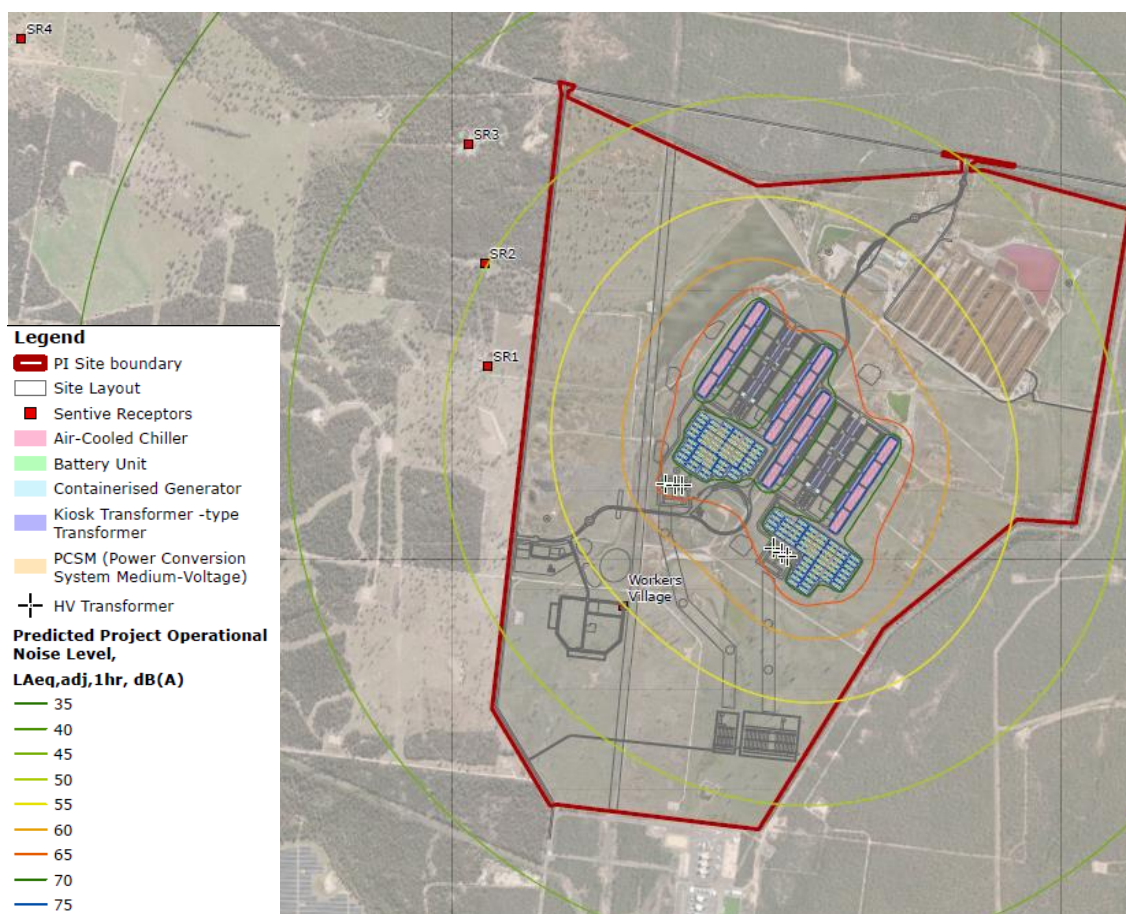
For further details, refer to **Appendix L**.

6.8 Noise Impact Assessment

Noise Impact Assessment has been prepared by **ERM** and is included at **Appendix M**. The assessment establishes the existing acoustic environment surrounding the site, develops project-specific operational noise criteria in accordance with the *Environmental Protection (Noise) Policy 2019*. Operational noise modelling considers normal operation, generator maintenance and emergency operation, together with potential construction noise and vibration impacts.

The is based on the current concept design and preliminary sound power data for the principal noise-generating equipment associated with the Research & Technology Industry and ancillary battery storage infrastructure. The assessment also accounts for existing, approved and reasonably foreseeable industrial and energy developments within the surrounding locality. The noise contribution from the proposed Grahams Road battery storage facility was found to be negligible relative to noise generated by the proposal. **Figure 24** illustrates the predicted noise contours for the worst-case daytime normal operating scenario.

Figure 24 – Predicted Operational Noise Contours



Source: ERM

6.8.1 Operational Noise Outcomes

The assessment indicates that the current concept design does not achieve the adopted project noise criteria at all identified sensitive receptors or the proposed Workforce Accommodation under the assessed operating scenarios. ERM attributes these exceedances to several factors, including:

- low existing background noise environment;
- number and distribution of proposed plant and equipment across the site;
- preliminary sound power levels adopted for key noise-generating equipment;
- the proximity of sensitive receptors to the site; and
- the conceptual nature of the current design, which does not yet incorporate the full suite of acoustic treatments and design refinements.

The assessment confirms that mitigation will be required before compliance can be demonstrated. Importantly, these outcomes are not considered to represent a fundamental constraint to the proposed development. Rather, they identify the areas where further acoustic optimisation, layout refinement and operational controls will need to be progressed through detailed design. It is also acknowledged that all sensitive receivers, with the exception of sensitive receiver 2 are owned by the same entity that owns the land that this development application is lodged over.

6.8.2 Detailed Design Response

The assessment identified a clear pathway to achieving compliance through an iterative acoustic design process. This will include consideration of:

- lower-noise chillers, battery units and associated plant;
- manufacturer-supplied attenuation packages, acoustic barriers and enclosures;
- revised equipment locations and site layout;
- use of buildings and other structures to provide acoustic shielding; and
- clearly defined operational controls during noise-sensitive periods.

Localised treatment close to the dominant noise sources is expected to be more effective than reliance on a site boundary barrier alone. Opportunities to relocate plant towards the centre of the site, orient noise-emitting equipment away from sensitive receptors and use the Research & Technology Industry buildings as acoustic shielding will therefore be tested as the design progresses.

The Workforce Accommodation will also be subject to a separate architectural acoustic assessment. The assessment outlines suitable internal noise levels are practicably achievable through appropriate building-envelope design, closed windows and mechanical ventilation or air-conditioning. Treatment of existing off-site dwellings would only be considered where source and pathway controls cannot reasonably achieve the required outcome and would require agreement with the relevant landowner.

Updated acoustic modelling will be undertaken before the equipment layout and plant specifications are finalised. This assessment will incorporate confirmed manufacturer sound data, the revised layout, proposed acoustic treatments and any operational controls relied upon to demonstrate compliance. Post-commissioning monitoring will then verify operational noise levels and the effectiveness of the adopted mitigation measures.

6.8.3 Construction Noise and Vibration

Construction activities with the potential to generate noise include site preparation, temporary compounds and batching facilities, installation of electrical infrastructure, piling and construction traffic. These impacts will be managed through standard construction hours and a **Construction Noise and Vibration Management Plan**, prepared before construction commences. The plan will identify relevant sensitive

receptors, proposed work activities, mitigation measures, monitoring procedures and processes for responding to complaints.

Construction vibration impacts are not expected due to the separation between the proposed works and surrounding sensitive receptors. Even higher-vibration activities, such as impact piling, are expected to occur well beyond the distances at which noticeable vibration impacts would ordinarily arise.

A detailed assessment of the existing acoustic environment, operational noise modelling, cumulative impact assessment and proposed mitigation measures as provided at **Appendix M**.

6.9 External Lighting Assessment

An External Lighting Assessment has been prepared by *HDR Pty Ltd (HDR)* and is included at **Appendix N**. The assessment establishes the external lighting strategy for the proposed development, including internal roads, pedestrian areas, car parks and the perimeter of the Research and Technology Industry. The lighting design has been developed to provide a safe and functional operating environment while minimising off-site light spill and impacts on the surrounding landscape.

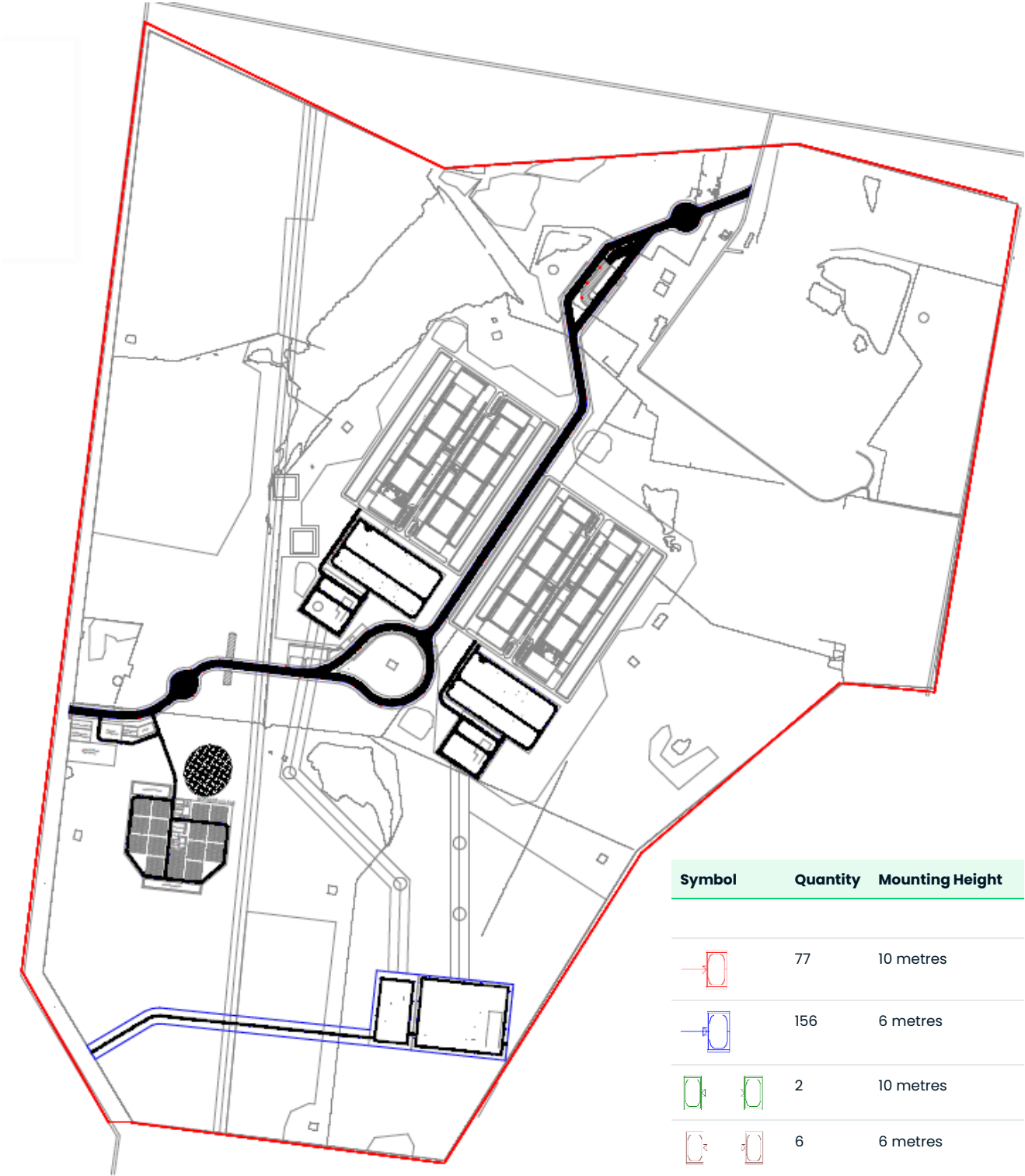
The proposed lighting network comprises pole-mounted LED luminaires distributed throughout the site to service major and minor internal roads, parking areas and pedestrian circulation routes. The layout has been designed to achieve the required operational lighting levels while optimising pole spacing to reduce the overall number of fittings across the development. **Figures 25 and 26** illustrate the proposed lighting arrangement and luminaire schedule.

A key objective of the lighting design is to minimise off-site impacts. All luminaires have been selected to achieve **0% Upward Light Output Ratio (ULOR)**, with fittings oriented to reduce light spill and sky glow beyond the site. The proposed luminaires are also certified as Australia Dark Sky compliant, incorporating replaceable LED modules and surge protection suitable for long-term operation.

The assessment concludes that the proposed lighting strategy is capable of delivering appropriate safety, accessibility and operational outcomes while limiting adverse lighting effects on adjoining land. The detailed lighting design will continue to be refined during the detailed design phase in accordance with the principles and recommendations outlined in the assessment.

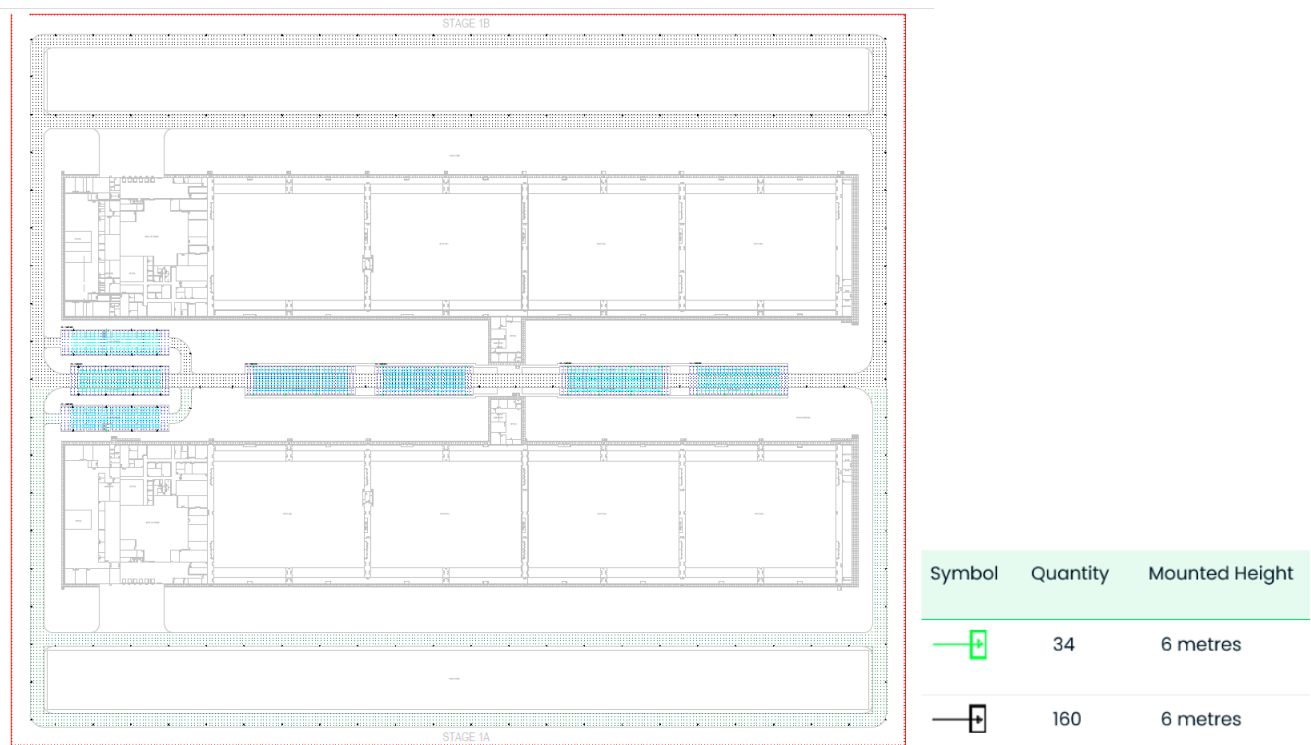
For further information, refer to **Appendix N**.

Figure 25 – External Lighting Schedule



Source: HDR

Figure 26 –Luminare Schedule (Internal Data Room & Car Parks)



Source: HDR

6.10 Waste Management Plan

A Waste Management Plan has been prepared by *Urbis* and is included at **Appendix O**. The report establishes the waste management strategy for the proposed development, including waste generation, storage requirements, collection arrangements and servicing. The assessment has been prepared for the ultimate development across Stages 1A, 1B, 2A and 2B, recognising that detailed design for each stage will continue to be refined through the staged development process.

6.10.1 Waste Generation and Storage

The assessment adopts waste generation rates for both the office and Research and Technology Industry components of the development to determine the required storage capacity for operational waste streams. These include general waste, commingled recycling, cardboard, food organics, soft plastics and electronic waste.

To service the staged development, dedicated waste management facilities are proposed within each development stage. Four waste rooms are distributed across the site, with each stage incorporating dedicated storage areas sized to accommodate the anticipated waste volumes generated by that stage.

The assessment confirms that the proposed waste rooms provide a combined storage area of approximately **216m²**, substantially exceeding the calculated minimum requirement of approximately **94m²**, allowing sufficient capacity for operational waste storage together with bulky waste where required.

6.10.2 Collection and Servicing

All waste collections will occur wholly within the site by a private waste contractor. Collection vehicles will access the development via *Grahams Road*, utilise the internal road network and undertake collections directly from the designated waste management areas before exiting the site without requiring bins to be presented to the public road.

The servicing strategy has been designed for collection by an 8.5m Medium Rigid Vehicle (MRV) or smaller, with a minimum clearance of 4.0m provided along the vehicle path. Swept path analysis has been undertaken to confirm that collection vehicles can safely access, manoeuvre within and exit the waste servicing areas. Building management will coordinate waste collections to ensure safe access for servicing vehicles and operational efficiency.

The Waste Management Plan concludes that the proposed waste storage, collection and servicing arrangements provide sufficient capacity to accommodate the operational requirements of the development while maintaining all waste handling and collection activities within the site boundary. Detailed waste management arrangements for each stage will continue to be refined as part of the detailed design process.

6.11 Economic Benefit Assessment

An Economic Benefits Assessment has been prepared by *Urbis* and is included at **Appendix P**. The assessment considers the economic contribution of the proposed development during both the construction and operational phases, including employment generation, Gross Value Added and the broader economic benefits to the Western Downs region and Queensland.

The assessment concludes that the proposed development will deliver significant economic benefits through major capital investment, employment creation and ongoing business activity. During the approximately four-year construction period, the project is expected to support around **1,500 direct on-site construction jobs**, generating substantial flow-on economic activity across the broader Queensland economy.

Upon full operation, the development is expected to support approximately **1,400 direct on-site jobs**, with the ongoing operation and maintenance of the facility contributing an estimated **\$760 million in annual GVA** to the Queensland economy. In addition to these direct economic benefits, the assessment identifies broader opportunities to strengthen the regional workforce through specialised technical employment, support skills development, attract complementary investment and enhance high-capacity digital infrastructure within the Western Downs region.

Further detail regarding the assessment methodology, economic modelling and projected employment outcomes is provided at **Appendix P**.

7 **Community and Stakeholder Engagement**

7 Community and Stakeholder Engagement

7.1 Engagement Approach

The applicant has developed a communication and engagement strategy to support the proposed Research and Technology Industry, Workforce Accommodation and High Impact Industry at 1933 Dalby-Kogan Road, Kogan, within the Western Downs Regional Council area. Largely, the engagement approach is designed to inform community and stakeholders about the proposal, provide opportunities for feedback, and support transparent and consistent communication throughout the planning process.

The engagement framework is informed by the International Association for Public Participation (IAP2) Public Participation Spectrum and the Queensland Government's community engagement principles. Given the project's regional location, separation from sensitive urban uses and anticipated low level of direct impact on most stakeholders, the engagement focus primarily aligns with the "Inform" and "Consult" levels of participation. This approach prioritises clear and accessible information sharing, with targeted consultation on matters of community interest and local benefit.

The engagement is guided by the following principles, ensuring it remains:

- **Inclusive** by using multiple engagement channels to reach a broad cross-section of the community,
- **Accessible** by providing clear, plain-English information in both digital and non-digital formats,
- **Community-focused** by recognising the Western Downs' strong local identity, agricultural and energy industry experience,
- **Timely** by initiating regular communication and responding promptly to enquiries and feedback,
- **Flexible** by adapting engagement methods in response to stakeholder needs and project progression.

Stakeholder identification and analysis form a key component of this methodology. Stakeholders include Western Downs local councillors, State and Federal representatives, regional economic organisations, adjacent landowners, local businesses, community groups, education providers and the broader Dalby and Kogan communities. A detailed stakeholder matrix and risk assessment informs the engagement approach, communication priorities and mitigation strategies.

A range of communication and consultation activities are proposed, including:

- Direct introductory correspondence with adjacent landowners and sensitive receivers,
- A dedicated project email address and website,
- Community newsletters along with letters to immediate neighbours,
- Individual briefings and targeted stakeholder meetings,
- Ongoing engagement with elected representatives and regional organisations,
- Social media and community sentiment monitoring,
- Responsive engagement activities, where requested by stakeholders.

The engagement strategy recognises key community interests including water efficiency, energy demand, road impacts, housing supply, employment opportunities, environmental management and long-term regional benefits. Key messaging focuses on the project's proposed contributions to economic diversification, digital infrastructure investment, local employment generation, infrastructure upgrades and sustainability outcomes, while also addressing potential impacts and mitigation measures.

Overall, the engagement process sits alongside yet separate to the public notification process undertaken as part of the development application. It aims to build informed community awareness, establish constructive stakeholder relationships, minimise misinformation, and support a transparent planning and assessment process for the proposed development

8 Conclusion

8 Conclusion

This report has been prepared in support of a Development Application seeking a Development Permit for a Material Change of Use for a Research and Technology Industry, Workforce Accommodation and High Impact Industry (Concrete Batching Plant), together with associated Operational Works, at 1933 Dalby–Kogan Road, Dalby.

The proposed Western Downs Digital Park represents a nationally significant digital infrastructure project that will establish Australia's largest data centre campus. The proposal has been strategically planned to leverage the Western Downs Region's established energy infrastructure, significant transmission network capacity and the site's location within the Kogan East Special Industry Area. In doing so, the development will reinforce the region's role as a key energy and technology hub while delivering substantial economic, employment and infrastructure benefits at a regional, state and national level.

This report demonstrates that the proposal has been comprehensively assessed against the relevant provisions of the *Planning Act 2016*, the *Planning Regulation 2017* and the Western Downs Planning Scheme 2017. While the proposal introduces land uses within the Rural Zone that would not ordinarily be anticipated and includes limited variations relating to building height and operational noise, these matters have been thoroughly considered and are appropriately justified having regard to the strategic intent of the Kogan East Special Industry Area, the exceptional characteristics of the site and the broader public interest served by the proposal.

The proposal achieves a high level of compliance with the Strategic Framework and relevant assessment benchmarks. The development is appropriately located within an established energy and industrial precinct, maintains appropriate separation from sensitive land uses, and incorporates significant setbacks, landscaping and environmental management measures to minimise potential off-site impacts. Comprehensive specialist investigations have confirmed that the proposal can be appropriately serviced and managed, with suitable mitigation measures addressing transport, infrastructure servicing, stormwater, ecology, fire safety, air quality, noise, lighting, waste management and environmental matters.

The proposal will deliver significant public benefits, including:

- establishment of a nationally significant digital infrastructure asset supporting Australia's growing digital economy;
- approximately \$31.9 billion in private capital investment;
- substantial employment opportunities during both construction and operation;
- diversification of the Western Downs economy through investment in advanced technology industries;
- efficient utilisation of existing regional energy infrastructure; and
- ongoing opportunities for local businesses, contractors and regional supply chains.

The proposed development represents an appropriate, orderly and strategically significant planning outcome. The proposal is supported by a comprehensive body of technical evidence demonstrating that potential impacts can be appropriately avoided, managed or mitigated, whilst delivering significant economic, employment and community benefits that strongly align with the strategic objectives of the planning framework.

Accordingly, having regard to the relevant assessment benchmarks, the supporting specialist investigations and the broader public interest, it is respectfully submitted that the proposed development warrants approval, subject to reasonable and relevant conditions.

Disclaimer

This report is dated 14 August 2026 and incorporates information and events up to that date only and excludes any information arising, or event occurring, after that date which may affect the validity of Urbis Ltd (**Urbis**) opinion in this report. Urbis prepared this report on the instructions, and for the benefit only, of WDDP Pty Ltd (**Instructing Party**) for the purpose of Town Planning Report (**Purpose**) and not for any other purpose or use. To the extent permitted by applicable law, Urbis expressly disclaims all liability, whether direct or indirect, to the Instructing Party which relies or purports to rely on this report for any purpose other than the Purpose, and to any other person which relies or purports to rely on this report for any purpose whatsoever (including the Purpose).

In preparing this report, Urbis was required to make judgements which may be affected by unforeseen future events, the likelihood and effects of which are not capable of precise assessment.

All surveys, forecasts, projections and recommendations contained in or associated with this report are made in good faith and on the basis of information supplied to Urbis at the date of this report, and upon which Urbis relied. Achievement of the projections and budgets set out in this report will depend, among other things, on the actions of others over which Urbis has no control.

In preparing this report, Urbis may rely on or refer to documents in a language other than English, which Urbis may arrange to be translated. Urbis is not responsible for the accuracy or completeness of such translations and disclaims any liability for any statement or opinion made in this report being inaccurate or incomplete arising from such translations.

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This report has been prepared with due care and diligence by Urbis and the statements and opinions given by Urbis in this report are given in good faith and in the reasonable belief that they are correct and not misleading, subject to the limitations above.

Appendix A – DA Form 1

Appendix B – Title Search and Landowner's Consent

Appendix C – Western Downs Planning Scheme 2017 Benchmark Assessments

Appendix D – Architectural Drawings

Appendix E – Landscape Plans

Appendix F – Traffic, Transport and Access Assessment

Appendix G – Stormwater Management Plan

Appendix H – Infrastructure Services Report

Appendix I – Fire Safety Concept Strategy

Appendix J – Environmental Assessment Report

Appendix K – EPBC Self-Assessment Report

Appendix L – Air Quality Assessment

Appendix M – Noise Impact Assessment

Appendix N – External Lighting Assessment

Appendix O – Waste Management Plan

Appendix P – Economic Benefit Assessment



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and communities
for a better future.**

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MATERIAL CHANGE OF USE

Recvd Date: 25/08/2026

Application: 030.2026.513.001

Customer: NAR No. 812575 Urbis Pty Ltd

Property: A41920

Subject: LG7.6.1

Precis: Corrected References Infrastructure Services Report Development

Application Material Change Use Lot 125 DY316 Emt C SP118043 A

RP119876 1933 Dalby-Kogan Road Dalby Western Downs Digital Park Pty
Ltd care Urbis Ltd Wambo Cattle Company Pty Ltd

Task Action: FYI only Planning, NFA required (Dominic Bradley, as assessing officer, included in email communication)



Infrastructure Services Report

Western Downs Digital Park

1933 Dalby-Kogan Road, Kogan, QLD 4406

Date: 25/08/26

Issue: 03

Reference: 10463553

Status: For Information

Prepared by: JL/NM

Date: 16/07/26

Edited by: NM

Date: 25/08/26

Authorised by: Naresh Mamtani

Date: 25/08/26

Issuing office: Sydney, AUS

DOCUMENT CONTROL

Issue	Date	Status	HDR Author ([XX/XX/XX].[Initials])	HDR Approval ([XX/XX/XX].[Initials])	Notes
01	24/07/2026	01	JL/NM	NM	For comments
02	18/08/2026	02	JL/NM	NM	For comments
03	25/08/2026	03	JL/NM	NM	For comments
04					
05					

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1.0 EXECUTIVE SUMMARY

This Infrastructure Services Report has been prepared by HDR to accompany a State Significant Development Application (DA) for the construction new data centre facility on Lot 125 on DY316 adjacent to the existing Wambo Feedlot in the Western Downs Region of Queensland in support of Urbis's Development Application for Project Western Downs Digital Park.

This report outlines the incoming services required, with respect to capacity, availability and connections to facilitate the development.

The proposed data centre development works specific to infrastructure include:

- Two customer-owned switching stations (275 kV and 330 kV). Site reticulation will consist of two double-circuit 330 kV overhead lines and a double-circuit 275 kV overhead line connecting the switching stations to the Level 2 substations, with 33 kV underground cabling distributing supply from each substation through its PCS/BESS compound to the data buildings.
- New pit and pipe system to facilitate incoming communication services pathway via diverse underground routes.
- Water & Sewer strategy for the site.

This report concludes that the proposed data centre development is suitable and warrants approval.

2.0 SITE AND LOCAL CONTEXT

The site is located at 1933 Dalby–Kogan Road, Dalby, and is formally described as Lot 125 on DY316. The site comprises a single freehold parcel of approximately 725.5 hectares (ha) with frontages to Dalby–Kogan Road along the northern boundary and Grahams Road along the eastern boundary. The site is located approximately 37km north-west of Dalby Town Centre.

The site is held under private tenure and is currently used for rural purposes comprising a cattle feedlot. The site contains an existing approval for a feedlot (known as 'Wambo Feedlot') (Council Ref.: DEV/2004/087). The approved feedlot has a capacity of 24,000 head of cattle. The feedlot and associated ancillary infrastructure are primarily located within the north-eastern portion of the site and include administrative offices, rural worker accommodation and effluent treatment infrastructure.

The site (as shown in Figure 1) is adjacent to existing electricity supply network infrastructure including the Alinta Braemar 1 and 2 Gas Power Stations, the Origin Energy Darling Downs Gas Power Station, the Powerlink's 330kV Braemar Substation and APA's Darling Downs Solar Farm.



Source: Nearmap, 2025

Figure 1 – Site Aerial

The site benefits from its strategic location within an established energy corridor, with access to existing transmission infrastructure and sufficient land area to accommodate large-scale, staged infrastructure development.

Existing irrigation areas associated with water reuse and waste treatment are centrally located on the site. The land is predominantly cleared of native vegetation and contains various existing gas well installations.

3.0 PROJECT DESCRIPTION

The proposed development, known as Western Downs Digital Park, comprises a large data centre supported by associated and ancillary infrastructure and land uses. At completion, the data centre will provide approximately 2,160 MVA total capacity, resulting in the data centre being the largest of its kind in Australia. The development will position Queensland, and the Western Downs region at the forefront of the evolving digital economy. An outlined of the Master Plan for the site is provided in Figure 1 below.



Figure 2 - Proposed view of the development looking north

The proposed development consists of the following elements:

- Four large data centre buildings, constructed over four development stages (Stages 1A, 1B, 2A and 2B), arranged centrally within the site.
- A secure operational campus incorporating controlled access points, boom gates, guard houses, perimeter fencing surrounding the operational areas.
- Components of associated and ancillary infrastructure to support an uninterrupted power supply to the proposed development, including:
 - Electrical infrastructure, including substations, switching yards and transmission lines to provide power to the proposed development;
 - Battery energy storage systems which provide for the storage of electricity to provide a source of power to the proposed development;
- A workers accommodation area which is proposed to provide accommodation for up to 1,500 construction workers during the construction of all stages of the proposed development. In addition to the residential accommodation, areas providing indoor and outdoor recreation and amenities will be provided given the scale of the workforce and the location related to Dalby. It is noted that whilst the location and layout of the accommodation have been shown in the drawings for the proposed development, there is anticipated to be some changes as a result of future design development.

4.0 INCOMING SERVICES

The incoming services outlined within this report include:

- Electricity
- Communication Services
- Potable Water
- Fire Water Infrastructure
- Sewerage
- Gas
- Stormwater

4.1 ELECTRICITY

Based on information gathered from AEMO shown in 3, the Braemar Transmission Substation is located approximately 600m south of the proposed site and operates at 275kV and 330kV, providing high-voltage transmission – this would be the preferred and closest connection point for the proposed data centre facility.

There are 275kV transmission lines that run approximately 1000m away from the northeast and approximately 400m and 180m away from the south of the site. A 275kV line also traverses the site near western boundary, as shown in the figure below. Additionally, there are 330kV transmission lines approximately 600m away from the south side of the proposed site.

Several gas power plants are located in the vicinity to the proposed site. Daandine Power Station is located approximately 1.6km away of the west of the site, Braemar Gas Power Station, Braemar 2 Gas Power Station and Darling Downs Gas Power Stations are located south of the site from approximately 50m, 400m and 900m distances, respectively.

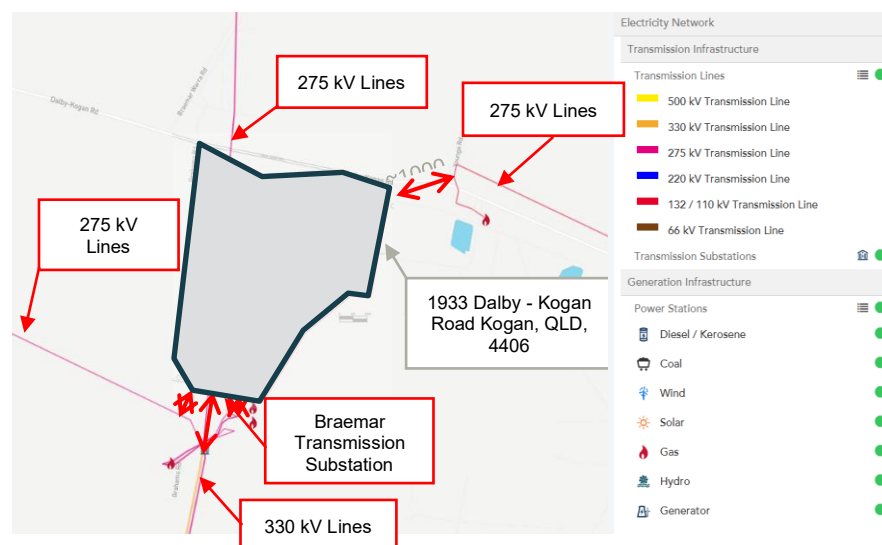


Figure 3 – Energy assets around 1933 Dalby – Kogan Road, QLD, 4406 (Source: AEMO)

There are medium-voltage (33kV and 12.7kV) electricity supplies and distribution substations operated by Ergon Energy in the vicinity of the site as shown in figure 4. These existing distribution substations typically have capacities up to 200kVA.

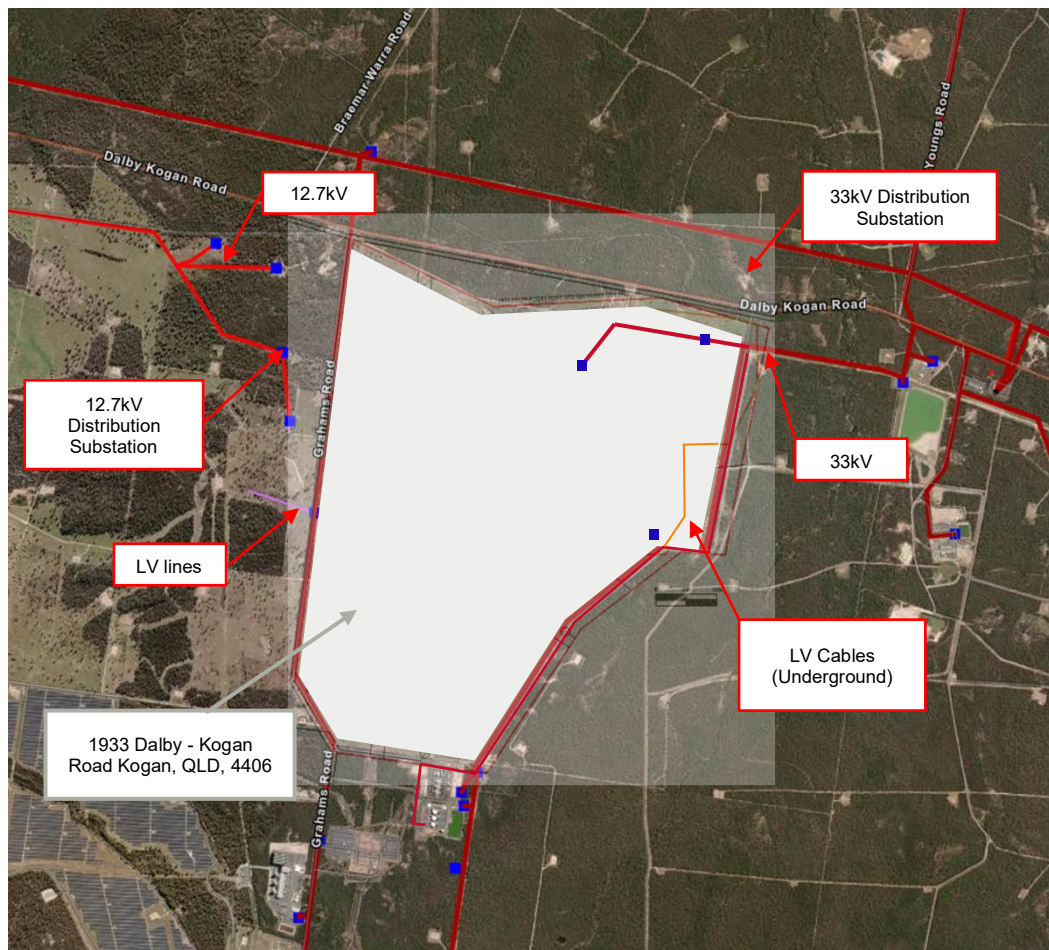


Figure 4 - Energy assets around 1933 Dalby – Kogan Road, QLD, 4406 (Source: Arcgis & Ergon Energy BYDA Response)

The electrical high voltage design for Western Downs Digital Park aims to provide adequate capacity and reliability for the maximum demand required of four data buildings, each with a total electrical demand of 540 MVA. Across all four buildings, the campus reaches approximately 2,160 MVA of total facility demand.

The high voltage system is designed to meet the maximum demand while maintaining N+1 redundancy continuously from the switching stations downwards across the site. A dual power conversion and battery energy storage system is integrated between each substation and the data buildings, providing two hours of autonomy per stage, decoupling the load from the network, and allowing the storage to participate in network services.

The campus will be supplied at two voltages i.e., 275kV and 330kV, through two customer-owned switching stations. The first is the 275kV switching station, supplied from two feeders from a power station, two from the Braemar 274kV substation (Powerlink), and five from the Darling Downs Power Station. The second station will operate at 330kV, supplied from the Braemar 330kV substation (Powerlink) via two feeders. The network connection points with Powerlink are located at the incoming Powerlink transmission feeders, at 275kV and 330kV, with revenue metering at the respective transmission voltages.

The campus is delivered in three stages, each establishing a Level 2 substation and the two data buildings it supplies, with a stage total facility demand of approximately 1,080MVA. Stage 1 is split into Stage 1A and Stage 1B to sequence the construction and energization of the 275kV switching station:

- Stage 1A establishes the 275kV switching station with its initial bays, supplied from the power station and Darling Downs Power Station, together with the together with the 275/33 kV substation and data building 1
- Stage 1B completes the remaining bays of the 275 kV switching station and connects them to the Braemar 275 kV substation and establishes data building 2. The design also allows for a contingency in which Stage 1 is supplied from a local gas source rather than the National Electricity Market.
- Stage 2 establishes the 330 kV switching station, connected to the Braemar 330 kV substation and serving the Stage 2 and Stage 3 loads, together with the first 330/33 kV substation and data buildings 3 and 4

Each stage is served by a dedicated Level 2 substation stepping the incoming transmission voltage down to 33 kV. Each substation comprises three 540 MVA three-winding transformers in a delta-star-star configuration. Redundancy is provided continuously from the switching stations downwards across the whole site, maintained through the transmission connections, the 33 kV substations, the power conversion and battery energy storage systems, and the data building distribution. The installed transformer capacity of 1,620 MVA per substation against a stage demand of approximately 1,080 MVA provides N+1 redundancy.

A battery energy storage system is integrated in series between each 33 kV incomer bus and the data building load through a dual power conversion arrangement. Each incomer bus connects to the grid-side power conversion system via 27 circuits (groups), each consisting of 8 strings. The grid-side converters feed the battery string, which in turn feeds the load-side converters. The outgoing feeders from the battery energy storage system feed the data building switch rooms, with four ring main unit rings distributed within each data building. The series battery energy storage system is multi-purpose: it supports energy arbitrage with the network and allows the campus to present a stable, near-constant load to the Powerlink network.

Stage 1 may be supplied either from the National Electricity Market via the Powerlink connection or from a local gas supply, with a gas well located on site. If Stage 1 is supplied via gas rather than the National Electricity Market, the battery energy storage system will not be constructed until Stage 2, and load fluctuation filtering will be provided in its place.

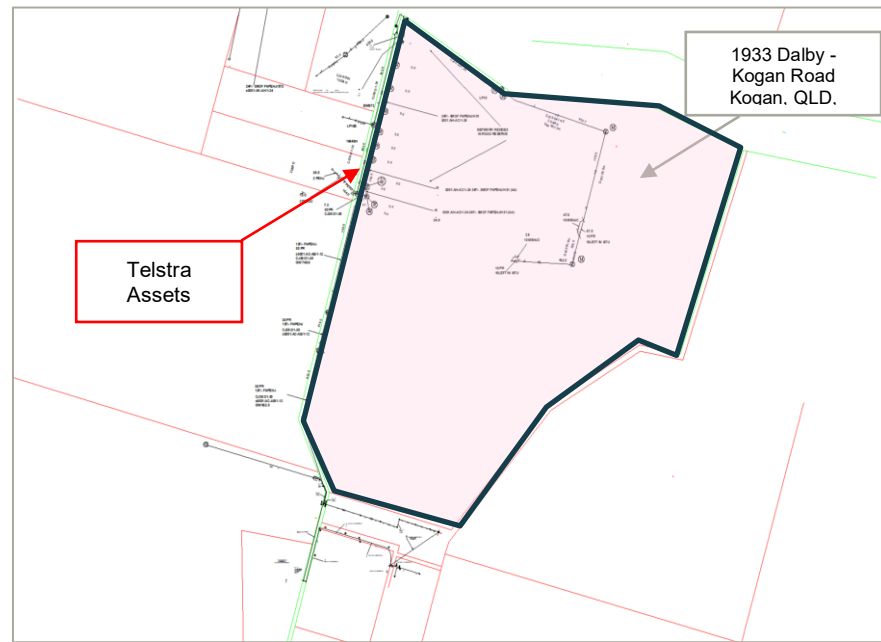
Site reticulation will consist of two double-circuit 330 kV overhead lines and a double-circuit 275 kV overhead line connecting the switching stations to the Level 2 substations, with 33 kV underground cabling distributing supply from each substation through its PCS/BESS compound to the data buildings.

Workers village – temporary connection 33kV on Grahams Road.

4.2 COMMUNICATION SERVICES

Initial BYDA enquiries indicate that several Telstra telecommunications assets are in the vicinity of the site as shown in figure 5. The preferred arrangement for fibre connection points to the site will be 4 geographically diverse paths with multiple carriers.

The diversely routed telecommunication service pathways within the building to the Telco/MDF rooms shall maintain a minimum of 20m separation along the entire route. The telco/ISP/Fibre service providers will likely provide connectivity to the fibre entry points to the site from public roads surrounding the site.



Source: BYDA Response - Telstra

Figure 5 – Telstra Assets around 1933 Dalby – Kogan Road, QLD, 4406 (Source: BYDA)

The proposed lead-in cabling strategy adopts a four-path interconnection model to provide diverse and scalable fibre connectivity for the campus, as shown in figure 6. The design incorporates both near-site and regional fibre paths to support future expansion and multiple fibre entry points. Telstra and QCN connections are treated as internal campus lead-ins due to their proximity to the site, whereas Dalby Exchange and Vocus/Nextgen provide external regional connectivity options.

- Telstra Interconnection Point – Provides the primary fibre connection for early-stage campus connectivity and integrates with the internal campus network.
- QCN / Powerlink Interconnection Point – Provides an additional near-site fibre path aligned with utility infrastructure to improve network diversity.
- Dalby Exchange Interconnection Point (with Telstra / Vocus / QCN optionality) – Provides a regional aggregation point that can potentially provide access to multiple carrier services and supports future network expansion
- Vocus / Nextgen Interconnection Point – Provides a long-haul connectivity option that improves carrier diversity and broader regional connectivity.

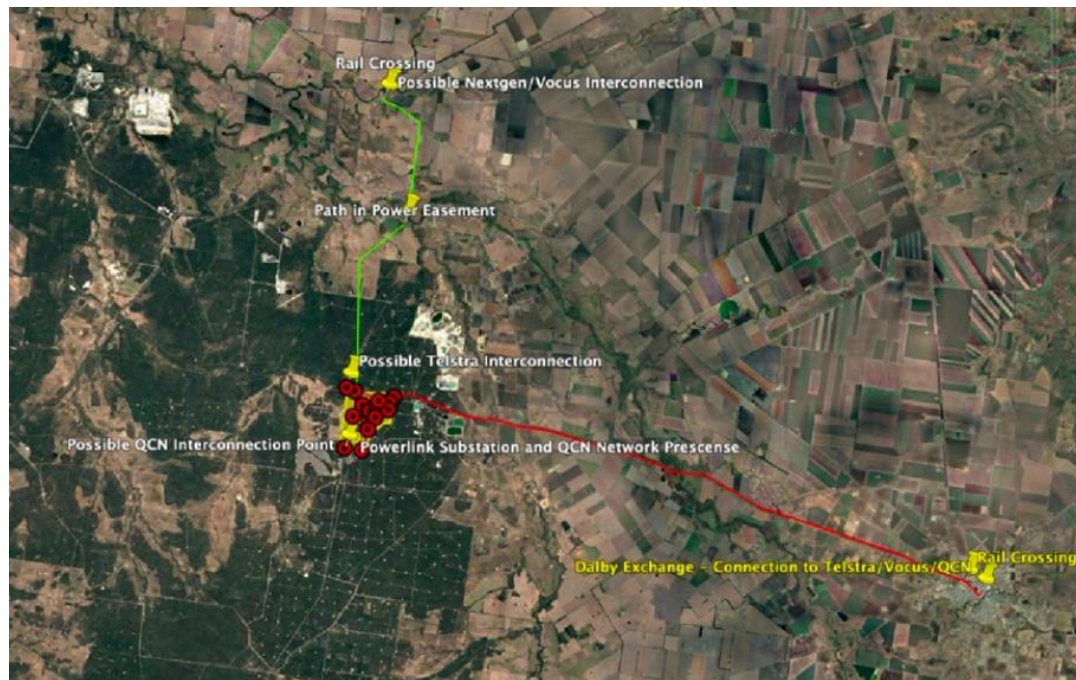


Figure 6 – Network design overview with proposed interconnection points

4.3 POTABLE AND NON-POTABLE WATER DEMAND

There is no reticulated (town) water network in the vicinity of the site, the nearest reticulated water infrastructure being approximately 15 km away. The development will therefore be serviced by an independent on-site water supply. This is a well-established servicing arrangement for developments of this nature in the region, and a number of practical and proven supply options are available to meet the development's water requirements for both the construction and operational phases.

Water demand arises from three sources: domestic potable demand (drinking, showering, hand basins and kitchen facilities); domestic non-potable demand (toilet and urinal flushing, washdown, general cleaning and landscape irrigation); and mechanical demand associated with the one-time commissioning (flushing and fill) of the cooling systems. Demand is highest during construction, when the workers' village accommodates a construction workforce of approximately 1,500 people, and reduces substantially to a modest permanent demand once construction is complete and the village is removed.

Project water demand is primarily associated with personnel onsite. There is no anticipated continuous water demand for equipment cooling; however, water will be required for the initial filling and commissioning of the closed-loop cooling systems as completed data halls are progressively commissioned. These requirements are considered separately in Section 3.5.

Other potential demands, including site irrigation, have not been included as potable water demands on the basis that treated recycled water is expected to be available from the wastewater treatment process.

The adopted personnel water demand assumptions are:

Personnel Category	Adopted Water Demand
Personnel residing onsite	215 L / person / day
Personnel residing offsite	54 L / person / day

Typical water demand for personnel residing in an accommodation camp range between approximately 150 and 200 L/person/day. An allowance of 215 L/person/day has been adopted for this assessment to provide an appropriate contingency.

Personnel residing offsite are assumed to consume approximately 25% of the water required by onsite personnel, resulting in an adopted demand of approximately 54 L/person/day.

The established Project water demand is expected to vary considerably throughout the Project lifecycle. Peak water demand occurs during periods of intensive construction activity and onsite accommodation, while long-term operational demand is significantly lower.

Based on the current personnel assumptions:

- peak water demand during construction is expected to exceed 300 kL/day;
- long-term operational water demand is significantly lower.

The significant variation between construction and operational demand is an important consideration in the selection, staging and sizing of water supply and treatment infrastructure.

The indicative water demand for Stage 1A (single building) is summarised below:

- Construction phase (construction site and workers' village): approximately 322 kL/day total (270 kL/day potable, 52 kL/day non-potable), or approximately 117 ML/year. An additional non-potable demand in the order of 200 kL/day may be required for construction dust suppression and cleaning.

CONSTRUCTION & WORKERS' VILLAGE						
Potable (DCW) and Non-Potable (NPW) shown separately and combined. Annual basis: 365 days/year.						
Component	Demand (kL/day)			Demand (ML/year)		
	Potable (DCW)	Non-Potable (NPW)	Total	Potable (DCW)	Non-Potable (NPW)	Total
Construction site (work shift)	36.0	31.5	67.5	13.1	11.5	24.6
Workers' village (after shift)	234.0	20.2	254.2	85.4	7.4	92.8
TOTAL	270.0	51.8	321.8	98.5	18.9	117.4

- Operational phase (one building, irrigation included in non-potable): approximately 16.5 kL/day total (10.6 kL/day potable, 5.9 kL/day non-potable), or approximately 4.8 ML/year.

BUILDING OPERATIONAL (one building)						
Irrigation included in NPW. Potable (DCW) and Non-Potable (NPW) shown separately and combined.						
Component	Demand (kL/day)			Demand (ML/year)		
	Potable (DCW)	Non-Potable (NPW)	Total	Potable (DCW)	Non-Potable (NPW)	Total
Operational building (incl. irrigation in NPW)	10.6	5.9	16.5	3.168	1.617	4.785

- Mechanical commissioning (one-time, per building): approximately 26.9 ML for the one-time flushing and fill of the cooling systems as each data building is brought online. This is a one-time commissioning requirement and not a recurring operational demand.

STAGE 1A — MECHANICAL WATER DEMAND SUMMARY (one building)			
Part	Volume (L)	Volume (kL)	Volume (ML)
DCW	25 969 833	25 969.8	25.970
DIW	890 000	890.0	0.890
STAGE 1A TOTAL (one-time)	26 859 833	26 859.8	26.860

The proposed solution has been developed in accordance with the Project's preferred water source hierarchy, prioritising CSG processed water during construction and rainwater harvesting during long-term operations.

The preferred solution incorporates:

- CSG processed water as the primary construction-phase water source;
- rainwater harvesting as the preferred operational water source;
- modular potable water treatment infrastructure;
- staged potable water storage;
- modular membrane bioreactor (MBR) wastewater treatment;
- maximum beneficial reuse of recycled water using HydraCool™ technology;
- irrigation and wet-weather storage for excess recycled water; and
- offsite management of treatment residuals where required.

The proposed solution provides a practical, flexible and scalable approach that minimises whole-of-life capital and operating costs while maintaining supply reliability throughout the Project lifecycle.

These options demonstrate that the development can be reliably serviced with potable and non-potable water. The final supply strategy will be confirmed during detailed design, including the mix of primary and supplementary sources, treatment where required, storage and any associated water entitlements, all of which are achievable through established processes.

4.4 FIRE WATER INFRASTRUCTURE

A combined Fire Hydrant and Sprinkler System is proposed to serve the new Data Centre Site. The combined system will primarily consist of:

- A full-capacity fire water storage system comprising two (2) half-capacity tanks, each with an effective capacity of 700,000 L.
- A water treatment and dosing facility to maintain fire water quality and to keep the water suitable for operation of the on-site fire pumps and for use by attending Fire Brigade personnel during firefighting operations.
- Two (2) fire pumps serving both the combined fire sprinkler and hydrant system, comprising one (1) primary full duty pump and one (1) standby full duty pump, each sized for 100% of the required system demand.

- A combined fire sprinkler and hydrant ring main supplying all on-site fire hydrant valves and sprinkler control valves, including preaction sprinkler control valves.
- A combined fire sprinkler and hydrant booster valve assembly, incorporating two (2) hardstand areas measuring 18 m × 6 m for Fire Brigade appliances. The booster valve assembly will need to be located adjacent to the fire water tanks and fire pump installation near the site boundary and principal vehicular entrance to the Data Centre site.
- A full backup water storage system comprising two (2) half-capacity tanks, each with an effective capacity of 700,000 L. These tanks will be used to refill the primary fire water storage system following operation. The backup tanks are required as there is no reticulated Town Main water supply available in the vicinity of the site. The site is located approximately 15 km from the nearest reticulated water infrastructure, and no connection to an existing potable water main is available.
- The backup water tanks will need to be refilled by a water tanker service provider following any drawdown of stored water.

A Fire Hydrant System is proposed to serve the Workers Village site. The fire hydrant system will primarily consist of:

- A full-capacity fire water storage system comprising two (2) half-capacity tanks, each with an effective capacity of 144,000 L.
- A water treatment and dosing facility to maintain fire water quality and to keep the water suitable for operation of the on-site fire pumps and for use by attending Fire Brigade personnel during firefighting operations.
- Two (2) fire pumps serving the combined fire hydrant system, comprising one (1) primary full duty pump and one (1) standby full duty pump, each sized for 100% of the required system demand.
- A reticulated main downstream of the fire hydrant pumps and booster valve assembly, supplying all on-site fire hydrant valves.
- A fire hydrant booster valve assembly, incorporating one (1) hardstand area measuring 18 m × 6 m for Fire Brigade appliances. The booster valve assembly will need to be located adjacent to the fire water tanks and fire pump installation near the site boundary and principal vehicular entrance to the Workers Village site.
- A full backup water storage system comprising two (2) half-capacity tanks, each with an effective capacity of 144,000 L. These tanks will be used to refill the primary fire water storage system following operation. The backup tanks are required as there is no reticulated Town Main water supply available in the vicinity of the site. The site is located approximately 15 km from the nearest reticulated water infrastructure, and no connection to an existing potable water main is available.
- The backup water tanks will need to be refilled by a water tanker service provider following any drawdown of stored water.

4.5 SEWER AND WASTEWATER

There is no reticulated sewerage network in the vicinity of the site. As with water supply, all wastewater generated by the development will be managed through an independent on-site arrangement, which is a well-established and readily achievable servicing solution for a development of this nature and location.

Wastewater is generated by the construction workforce and workers' village during construction, and by the operational staff once each building is commissioned, broadly in proportion to the potable water consumed. The development can be serviced through the following practical wastewater management options:

- Off-site removal – wastewater collected in on-site holding tanks and removed from site by a licensed liquid waste contractor for disposal at a licensed facility. This requires no on-site treatment and can be deployed from the commencement of works, providing a readily available solution for the construction phase and as a contingency arrangement.
- On-site wastewater treatment – treatment in an on-site sewage treatment plant, with the treated effluent disposed of by land application (subsurface irrigation to a dedicated land application area, or evaporation) and/or reused as non-potable water for toilet flushing, washdown and irrigation. This provides a sustainable long-term wastewater solution and reduces the development's reliance on imported water through effluent reuse.

An area will be reserved on the site for a future on-site wastewater treatment facility and its associated land application area, ensuring the long-term solution can be accommodated as the development proceeds. The treatment process, disposal and reuse arrangements and associated approvals will be confirmed during detailed design through established regulatory processes.

4.6 GAS

According to the QGC BYDA response, there are existing Drilled Wells, Gas Gathering Pipelines, Water Gathering Pipelines within the proposed land.

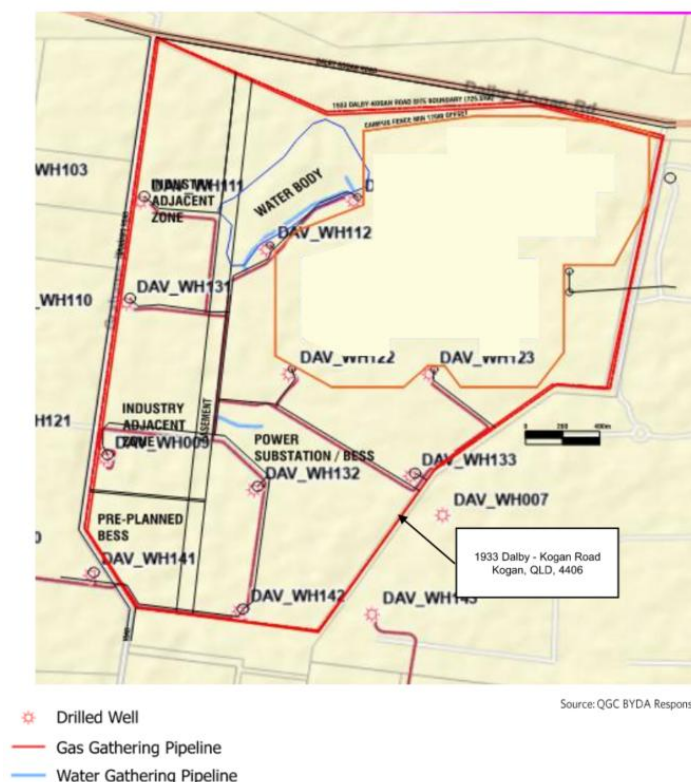


Figure 7 – QGC Assets around 1933 Dalby – Kogan Road, QLD, 4406 (Source: BYDA)

As per the APA Group Gas Transmission BYDA response, there is a high-pressure gas pipeline along the north boundary of the site as shown below.

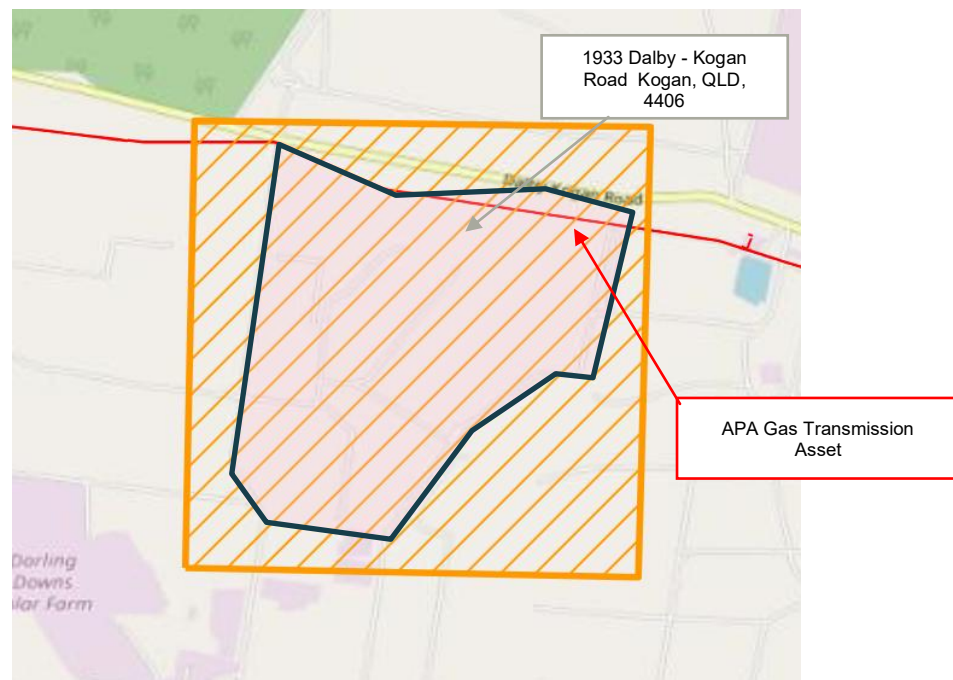
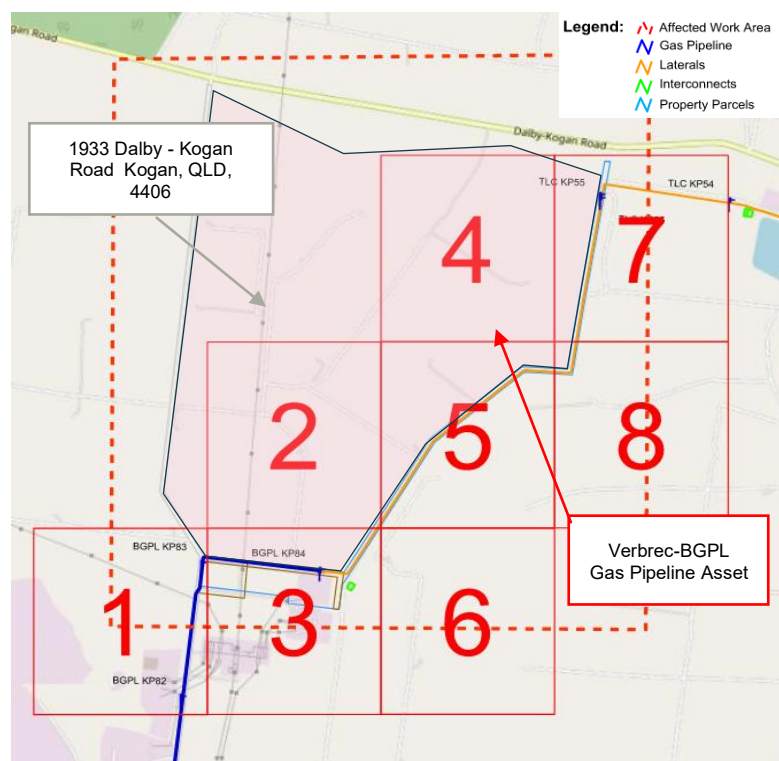


Figure 8 – APA Assets around 1933 Dalby – Kogan Road, QLD, 4406 (Source: BYDA)

According to the Verbrec-BPGL BYDA response, there is a gas property parcel including, gas pipeline, laterals, and interconnects along the east and south boundary of the site, which connect gas to the adjacent Alinta Energy Braemar Power Station.



Source: Verbrec-BPGL BYDA Response

Figure 9 – Verbrec-BPGL Assets around 1933 Dalby – Kogan Road, QLD, 4406 (Source: BYDA)

According to the Arrow Energy BYDA response, there are several gas pipelines close to the north boundary and east boundary of the site.

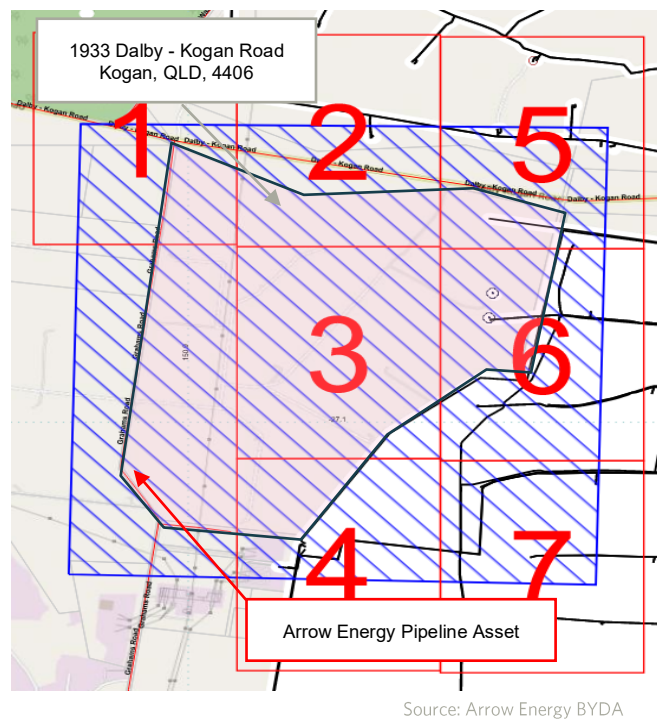


Figure 10 – Arrow Assets around 1933 Dalby – Kogan Road, QLD, 4406 (Source: BYDA)